

Signal and Systems Analysis

Homework Solutions

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Signal and Systems Analysis
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1 Homework Set 1

1.1 Problem 1

Please show all work with formal proofs.

- Using the basis $\{1, t, t^2\}$, defined for t values from -1 to 1 , find the best mean-square approximation of the function shown in Figure 1 (i.e. minimize the Euclidian norm of the difference between this function and its approximation). Is this basis orthogonal?
- Find the best linear mean-square approximation of the function in Figure 1 using the basis $\{1, t\}$. Is this basis orthogonal?
- Compare the errors of the two approximations in light of the fact that the second basis is a subset of the first.

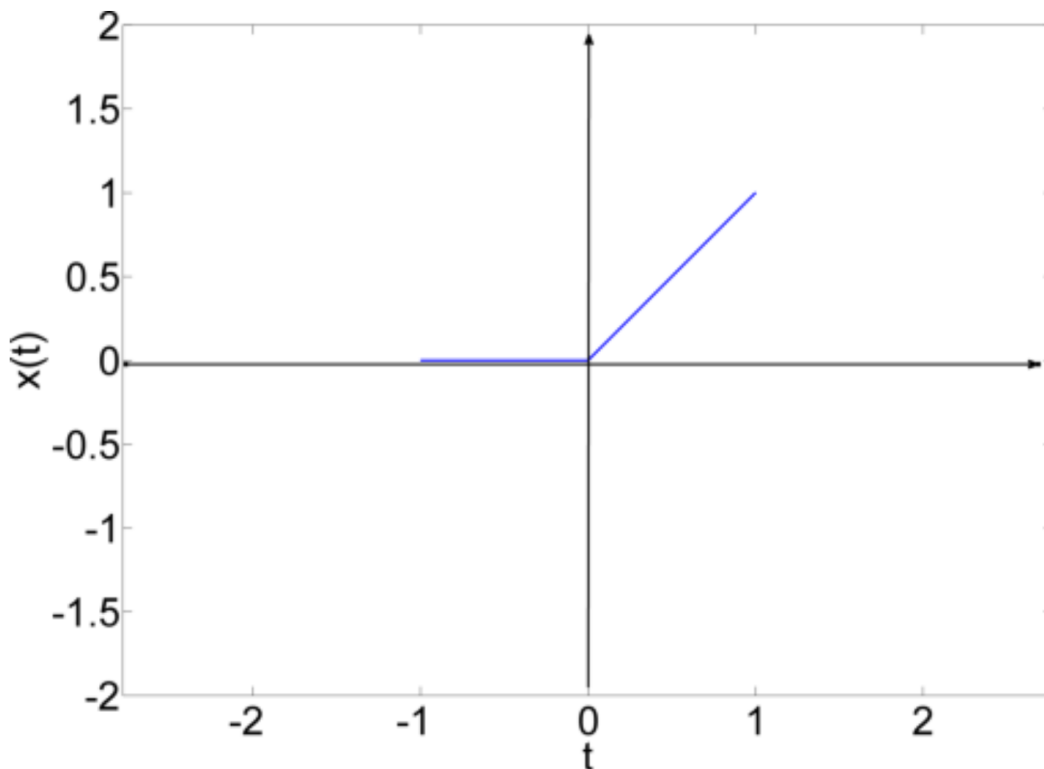


Figure 1

1.1.1 Solution

Let

$$x(t) = \begin{cases} 0, & -1 \leq t < 0, \\ t, & 0 \leq t \leq 1, \end{cases} \quad (1)$$

with inner product

$$\langle f, g \rangle = \int_{-1}^1 f(t)g(t) dt$$

and norm $\|f\|_2 = \sqrt{\langle f, f \rangle}$.

Theorem 1.1 (Quadratic Approximation). *Let $V = \text{span}\{1, t, t^2\} \subset L^2([-1, 1])$. The unique best mean-square approximant of x in V is*

$$p_V(t) = \frac{3}{32} + \frac{1}{2}t + \frac{15}{32}t^2. \quad (2)$$

Moreover, the set $\{1, t, t^2\}$ is not an orthogonal basis of V , and the mean-square error is

$$\|x - p_V\|_2^2 = \frac{1}{384}.$$

Proof. Denote $p_V(t) = a_0 + a_1t + a_2t^2$ and impose the projection conditions $\langle x - p_V, \phi \rangle = 0$ for each $\phi \in \{1, t, t^2\}$. The necessary inner products are

$$\begin{aligned} \langle 1, 1 \rangle &= 2, & \langle 1, t \rangle &= 0, & \langle 1, t^2 \rangle &= \frac{2}{3}, \\ \langle t, t \rangle &= \frac{2}{3}, & \langle t, t^2 \rangle &= 0, & \langle t^2, t^2 \rangle &= \frac{2}{5}, \\ \langle x, 1 \rangle &= \frac{1}{2}, & \langle x, t \rangle &= \frac{1}{3}, & \langle x, t^2 \rangle &= \frac{1}{4}. \end{aligned}$$

The normal equations $Ga = b$, with

$$G = \begin{bmatrix} 2 & 0 & \frac{2}{3} \\ 0 & \frac{2}{3} & 0 \\ \frac{2}{3} & 0 & \frac{2}{5} \end{bmatrix}, \quad b = \begin{bmatrix} \frac{1}{2} \\ \frac{1}{3} \\ \frac{1}{4} \end{bmatrix},$$

yield $a_1 = \frac{1}{2}$ and $(a_0, a_2) = (\frac{3}{32}, \frac{15}{32})$, giving (2). Since $\langle 1, t^2 \rangle = \frac{2}{3} \neq 0$, the basis is not orthogonal. The mean-square error is

$$\|x - p_V\|_2^2 = \|x\|_2^2 - a^\top b = \frac{1}{3} - \frac{127}{384} = \frac{1}{384}.$$

□

Theorem 1.2. *Let $W = \text{span}\{1, t\}$. The best mean-square approximant of x in W is*

$$p_W(t) = \frac{1}{4} + \frac{1}{2}t \quad (3)$$

the basis $\{1, t\}$ is orthogonal, and the mean-square error is

$$\|x - p_W\|_2^2 = \frac{1}{24}.$$

Proof. Set $p_W(t) = b_0 + b_1 t$ and impose $\langle x - p_W, \phi \rangle = 0$ for each $\phi \in \{1, t\}$. With

$$\begin{bmatrix} 2 & 0 \\ 0 & \frac{2}{3} \end{bmatrix} \begin{bmatrix} b_0 \\ b_1 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} \\ \frac{1}{3} \end{bmatrix},$$

we obtain $(b_0, b_1) = (\frac{1}{4}, \frac{1}{2})$, giving (3). Orthogonality follows from $\langle 1, t \rangle = 0$. The mean-square error is

$$\|x - p_W\|_2^2 = \|x\|_2^2 - b^\top \begin{bmatrix} b_0 \\ b_1 \end{bmatrix} = \frac{1}{3} - \frac{7}{24} = \frac{1}{24}.$$

□

Theorem 1.3. *Since $W \subset V$,*

$$\|x - p_V\|_2^2 \leq \|x - p_W\|_2^2.$$

For this problem,

$$\|x - p_V\|_2^2 = \frac{1}{384} < \|x - p_W\|_2^2 = \frac{1}{24}.$$

Proof. Since p_V is the orthogonal projection of x onto V , the residual $r = x - p_V$ satisfies $r \perp V$, and hence $r \perp (p_V - p_W)$. Thus,

$$x - p_W = r + (p_V - p_W)$$

and by the Pythagorean theorem,

$$\|x - p_W\|_2^2 = \|r\|_2^2 + \|p_V - p_W\|_2^2 \geq \|r\|_2^2 = \|x - p_V\|_2^2.$$

Note that strict inequality holds since $p_V \neq p_W$.

□

1.2 Problem 2

Please show all work with formal proofs.

- Using the basis $\{\sin(\pi t), \sin(2\pi t), \sin(4\pi t)\}$, defined for t from -1 to $+1$, find the best mean-square approximation of the function shown in Figure 2. Is this basis orthogonal?
- Find the best mean-square approximation of the function in Figure 2 using the function $\sin(2\pi t)$ (i.e. a basis with a single function). Compare the errors of the two approximations

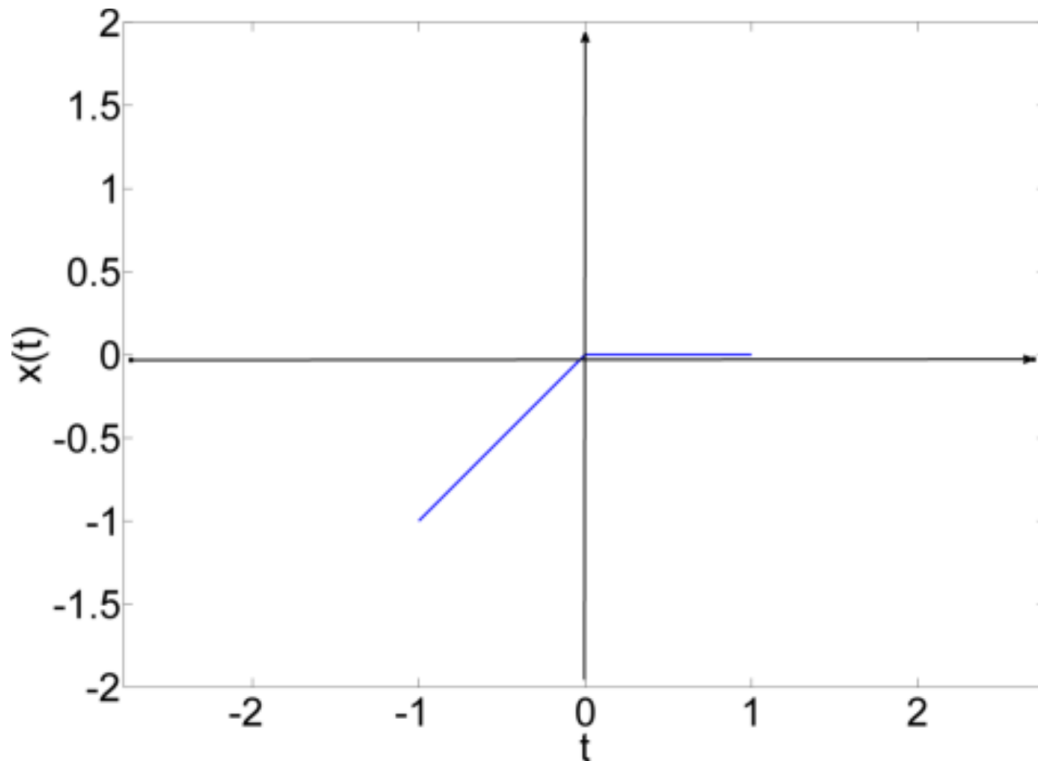


Figure 2

1.2.1 Solution

Let

$$x(t) = \begin{cases} t, & -1 \leq t \leq 0, \\ 0, & 0 < t \leq 1, \end{cases} \quad (4)$$

with inner product and norm

$$\langle f, g \rangle = \int_{-1}^1 f(t)g(t) dt, \quad \|f\|_2^2 = \langle f, f \rangle.$$

A direct computation gives

$$\|x\|_2^2 = \int_{-1}^0 t^2 dt = \frac{1}{3}. \quad (5)$$

1.2.1.1 Part A

Theorem 1.4. *The functions $\{\sin(\pi t), \sin(2\pi t), \sin(4\pi t)\}$ form an orthonormal basis of $V \subset L^2([-1, 1])$. The orthogonal projection of x onto V is*

$$P_V x(t) = \frac{1}{\pi} \sin(\pi t) - \frac{1}{2\pi} \sin(2\pi t) - \frac{1}{4\pi} \sin(4\pi t), \quad (6)$$

and the squared error is

$$\|x - P_V x\|_2^2 = \frac{1}{3} - \frac{21}{16\pi^2}. \quad (7)$$

Proof. For integers $m, n \geq 1$,

$$\int_{-1}^1 \sin(m\pi t) \sin(n\pi t) dt = \begin{cases} 1, & m = n, \\ 0, & m \neq n, \end{cases}$$

since $\int_{-1}^1 \sin^2(n\pi t) dt = 1$. Thus, these sines form an orthonormal basis of V . Now, since $x(t) = 0$ on $(0, 1]$,

$$\begin{aligned} \langle x, \sin(k\pi t) \rangle &= \int_{-1}^0 t \sin(k\pi t) dt \\ &= \left[-\frac{t \cos(k\pi t)}{k\pi} + \frac{\sin(k\pi t)}{(k\pi)^2} \right]_{-1}^0 \\ &= -\frac{(-1)^k}{k\pi} = \frac{(-1)^{k+1}}{k\pi}. \end{aligned}$$

Thus, $a_1 = \frac{1}{\pi}$, $a_2 = -\frac{1}{2\pi}$, and $a_4 = -\frac{1}{4\pi}$. Since the basis is orthonormal,

$$P_V x(t) = \sum_{k \in \{1, 2, 4\}} a_k \sin(k\pi t),$$

giving Equation (6). By Parseval's identity,

$$\|x - P_V x\|_2^2 = \|x\|_2^2 - \sum_{k \in \{1, 2, 4\}} a_k^2 = \frac{1}{3} - \frac{1}{\pi^2} \left(1 + \frac{1}{4} + \frac{1}{16} \right) = \frac{1}{3} - \frac{21}{16\pi^2}.$$

□

Lemma 1.5. *Let $\phi_1 = \sin(\pi t)$, $\phi_2 = \sin(2\pi t)$, $\phi_3 = \sin(4\pi t)$. The Gram matrix $G = (\langle \phi_i, \phi_j \rangle)_{i,j}$ equals the identity I_3 , so the normal equations*

$$Ga = b, \quad b_i = \langle x, \phi_i \rangle,$$

have the solution

$$a = \left(\frac{1}{\pi}, -\frac{1}{2\pi}, -\frac{1}{4\pi}\right)^T,$$

confirming (6) and (7).

Proof. Product-to-sum formulas show $\langle \phi_i, \phi_j \rangle = 0$ for $i \neq j$ and $\langle \phi_i, \phi_i \rangle = 1$. Thus, $G = I_3$, and the system $Ga = b$ reduces to $a = b$, recovering the coefficients from Theorem 1.4. $\square$

1.2.1.2 Part B

Theorem 1.6. *The orthogonal projection of x onto W is*

$$P_W x(t) = -\frac{1}{2\pi} \sin(2\pi t), \quad (8)$$

and the squared error is

$$\|x - P_W x\|_2^2 = \frac{1}{3} - \frac{1}{4\pi^2}. \quad (9)$$

Proof. Since $\sin(2\pi t)$ is unit-norm, its Fourier coefficient is $a_2 = \langle x, \sin(2\pi t) \rangle = -\frac{1}{2\pi}$, and $P_W x = a_2 \sin(2\pi t)$. By Parseval's identity,

$$\|x - P_W x\|_2^2 = \|x\|_2^2 - a_2^2 = \frac{1}{3} - \frac{1}{4\pi^2}.$$

$\square$

Theorem 1.7. *Since $W \subset V$ and $P_V x \neq P_W x$,*

$$\|x - P_V x\|_2^2 < \|x - P_W x\|_2^2. \quad (10)$$

Numerically,

$$\|x - P_V x\|_2^2 \approx 0.2003, \quad \|x - P_W x\|_2^2 \approx 0.3080.$$

Proof. For nested subspaces $W \subset V$ of a Hilbert space, $\|x - P_V x\|_2 \leq \|x - P_W x\|_2$, with equality only if $P_V x = P_W x$. From (6) and (8) we see that $P_V x$ contains nonzero $\sin(\pi t)$ and $\sin(4\pi t)$ components absent from $P_W x$, so equality cannot hold. Substituting $\pi^2 \approx 9.8696$ into (7)–(9) gives the numerical values. $\square$

2 Homework Set 2

2.1 Problem 1

Find the Fourier expansion of the following functions $x(t)$, defined over the interval $[0, 1]$.

1. $x(t) = \exp(-\alpha t)$
2. $x(t) = \sin(\pi t)$
3. $x(t) = \cos(\pi t)$

2.1.1 Solution

Proof. Let $x(t)$ be defined over $[0, 1]$ and extended periodically with period $T = 1$. Then, the complex Fourier series expansion is

$$x(t) = \sum_{n=-\infty}^{\infty} c_n e^{j2\pi n t}, \quad c_n = \int_0^1 x(t) e^{-j2\pi n t} dt.$$

(1) Consider $x(t) = e^{-\alpha t}$. Then,

$$\begin{aligned} c_n &= \int_0^1 e^{-\alpha t} e^{-j2\pi n t} dt = \int_0^1 e^{-(\alpha + j2\pi n)t} dt \\ &= \left[\frac{e^{-(\alpha + j2\pi n)t}}{-(\alpha + j2\pi n)} \right]_0^1 \\ &= \frac{1 - e^{-(\alpha + j2\pi n)}}{\alpha + j2\pi n}. \end{aligned}$$

(2) Using Euler's identity,

$$\sin(\pi t) = \frac{e^{j\pi t} - e^{-j\pi t}}{2j}.$$

Then,

$$\begin{aligned} c_n &= \int_0^1 \sin(\pi t) e^{-j2\pi n t} dt \\ &= \frac{1}{2j} \left[\int_0^1 e^{j\pi t} e^{-j2\pi n t} dt - \int_0^1 e^{-j\pi t} e^{-j2\pi n t} dt \right]. \end{aligned}$$

Only $n = \pm 1$ give nonzero values. A direct evaluation yields

$$c_{\pm 1} = \mp \frac{1}{2j}.$$

(3) Using Euler's identity,

$$\cos(\pi t) = \frac{e^{j\pi t} + e^{-j\pi t}}{2}.$$

Then,

$$\begin{aligned} c_n &= \int_0^1 \cos(\pi t) e^{-j2\pi n t} dt \\ &= \frac{1}{2} \left[\int_0^1 e^{j\pi t} e^{-j2\pi n t} dt + \int_0^1 e^{-j\pi t} e^{-j2\pi n t} dt \right]. \end{aligned}$$

Again only $n = \pm 1$ contribute, and computation gives

$$c_{\pm 1} = \frac{1}{2}.$$

□

2.2 Problem 2

find the Fourier Transforms of the following.

1. the complex exponential function: $x(t) = \exp(j\alpha t)$
2. the sinusoidal functions: $\cos(\alpha t)$ and $\sin(\alpha t)$
3. the pulse function: $p(t) = 1$ for all $|t| \leq W$ and 0 elsewhere
4. the periodic function: $x(t) = |\sin(\pi t)|$ for all $t \in (-\infty, \infty)$
5. the Comb function: $\text{comb}_T(t) = \sum \delta(t - mT)$, where T denotes the distance between successive delta functions and the summation is taken over all integers m

2.2.1 Solution

Proof. Let $\mathcal{F}\{x(t)\} = X(f)$ denote the continuous-time Fourier Transform.

$$X(f) = \int_{-\infty}^{\infty} x(t) e^{-j2\pi ft} dt. \quad (11)$$

Then, use standard transform pairs and properties.

(1) Using the Modulation Property.

$$\mathcal{F}\{e^{jat}\} = \delta(f - \frac{a}{2\pi}). \quad (12)$$

(2) Using Euler identities.

$$\begin{aligned} \cos(at) &= \frac{1}{2}(e^{jat} + e^{-jat}) \\ \Rightarrow \mathcal{F}\{\cos(at)\} &= \frac{1}{2}[\delta(f - \frac{a}{2\pi}) + \delta(f + \frac{a}{2\pi})] \\ \sin(at) &= \frac{1}{2j}(e^{jat} - e^{-jat}) \\ \Rightarrow \mathcal{F}\{\sin(at)\} &= \frac{1}{2j}[\delta(f - \frac{a}{2\pi}) - \delta(f + \frac{a}{2\pi})] \end{aligned}$$

(3) This is a rectangular pulse of width $2W$. The known result is provided below.

$$\mathcal{F}\{p(t)\} = 2W \text{sinc}(2Wf), \quad \text{where } \text{sinc}(x) = \frac{\sin(\pi x)}{\pi x}. \quad (13)$$

(4) Recognize this is periodic with period 1. Use the Fourier series, then transform.

- Compute Fourier series coefficients c_n on $[0, 1]$.
- FT: $X(f) = \sum_{n=-\infty}^{\infty} c_n \delta(f - n)$.

This results in line spectra at integer frequencies.

(5) The Poisson Summation Formula yields the following.

$$\mathcal{F}\{\text{comb}_T(t)\} = \frac{1}{T} \text{comb}_{1/T}(f). \quad (14)$$

Thus, the FT of a delta train is another delta train in frequency. □

2.3 Problem 3

Using Matlab, generate for $n = 1, \dots, 1024$ the discrete-time signals $x(n)$ and $y(n)$ given as follows.

$$x(n) = \exp\left(\frac{-n}{400}\left[\cos\left(\frac{n\pi}{64}\right) - \sin\left(\frac{n\pi}{8}\right)\right]\right)$$

$$y(n) = \cos\left(\frac{n^2\pi}{8192}\right) + \sin\left(\frac{n\pi}{32}\right)$$

Compute their discrete Fourier Transforms (DFT) and plot the resulting magnitude and phase of these DFTs. Discuss the results (e.g. what are their bandwidths, where are their resonant peaks etc.).

2.3.1 Solution

The computed data results and figures are shown below.

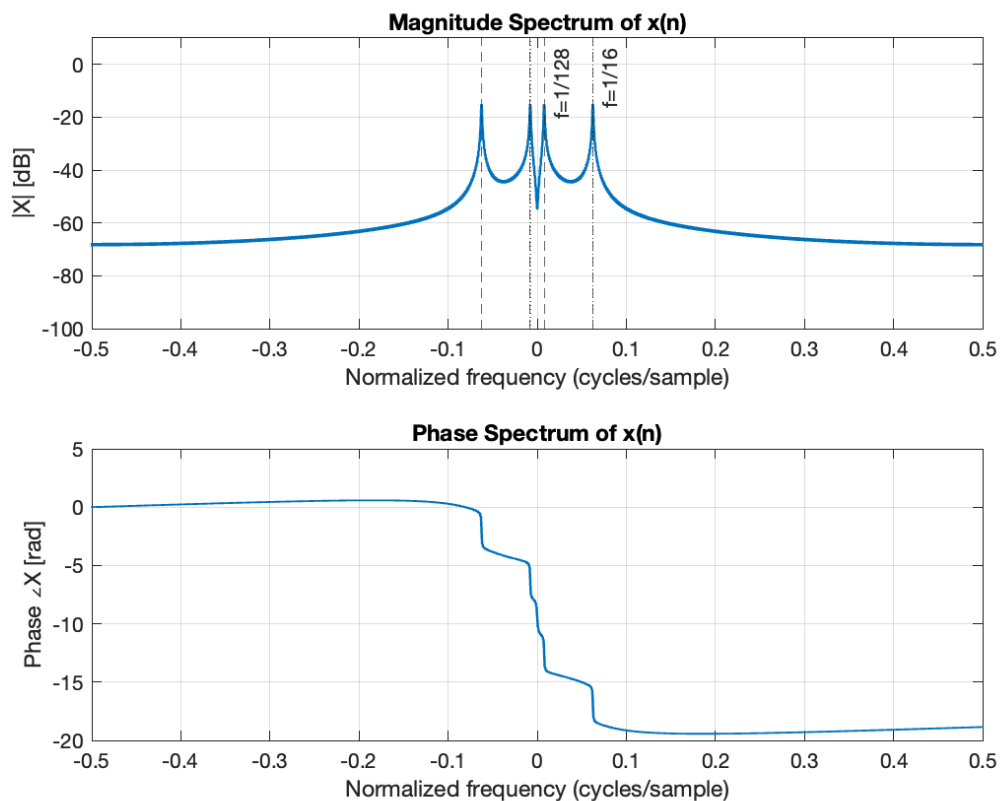


Figure 3: DFT magnitude and phase spectra of $x(n) = e^{-n/400} [\cos(n\pi/64) - \sin(n\pi/8)]$. Vertical dashed lines mark resonant frequencies at $f = 1/128$ and $f = 1/16$ cycles/sample, while dotted lines show the measured -3 dB bandwidth limits.

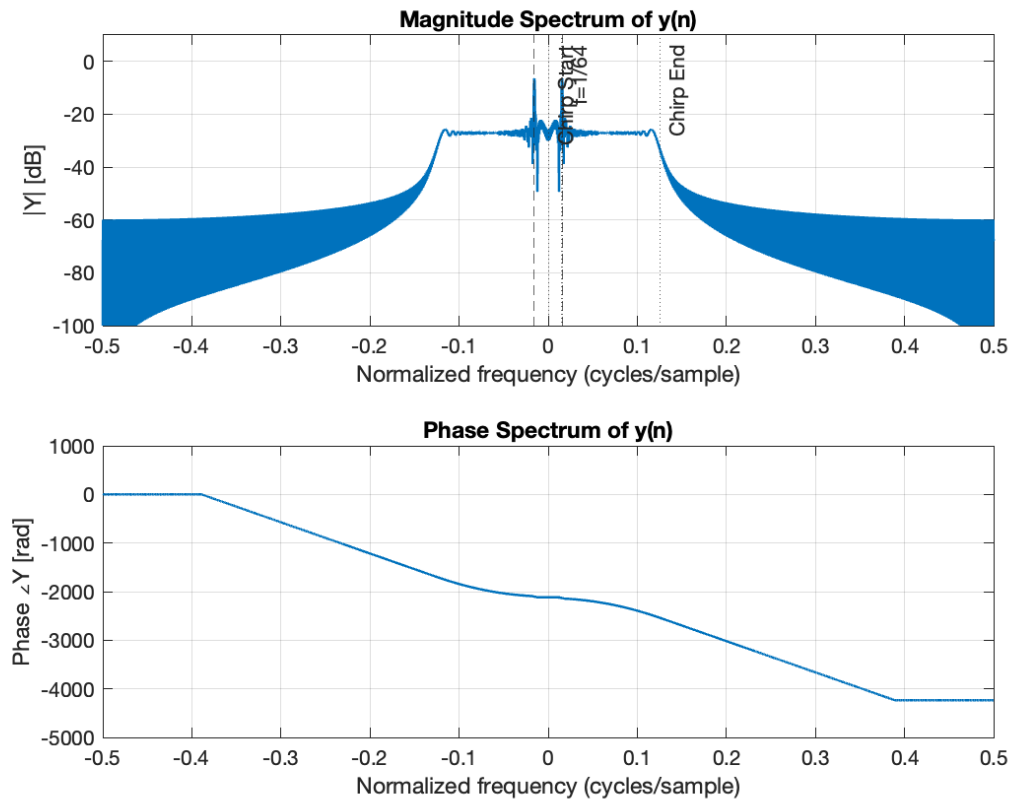


Figure 4: DFT magnitude and phase spectra of $y(n) = \cos\left(\frac{\pi n^2}{8192}\right) + \sin\left(\frac{n\pi}{32}\right)$. The broadband region corresponds to the chirp term sweeping from 0 to 0.125 cycles/sample, and the narrow spike near $f = 1/64$ arises from the fixed sinusoid.

Table 1: Measured -3 dB Bandwidths for $x(n)$ and $y(n)$

Signal	f_0	f_L	f_R	$\text{BW}_{-3\text{dB}}$	Peak [dB]	f_{peak}
$x(n)$	0.0078125	-0.00836	-0.00732	0.00104	-15.01	-0.00781
$x(n)$	0.0625	0.06198	0.06301	0.00104	-15.00	0.0625
$y(n)$	0.015625	0.01513	0.01600	0.00087	-6.38	0.01563

2.4 Problem 4

Prove the Fourier Transform properties (6)-(8) and (11)-(12) listed in the Class Notes Part 4.

2.4.1 Solution

Proof. Proving several canonical properties of the Fourier Transform using the standard definition.

$$\mathcal{F}\{x(t)\} = X(f) = \int_{-\infty}^{\infty} x(t)e^{-j2\pi ft} dt. \quad (15)$$

If $x_1(t) \leftrightarrow X_1(f)$ and $x_2(t) \leftrightarrow X_2(f)$, then,

$$\mathcal{F}\{ax_1(t) + bx_2(t)\} = aX_1(f) + bX_2(f)$$

since,

$$\begin{aligned} \mathcal{F}\{ax_1(t) + bx_2(t)\} &= \int_{-\infty}^{\infty} [ax_1(t) + bx_2(t)]e^{-j2\pi ft} dt \\ &= a \int_{-\infty}^{\infty} x_1(t)e^{-j2\pi ft} dt + b \int_{-\infty}^{\infty} x_2(t)e^{-j2\pi ft} dt \\ &= aX_1(f) + bX_2(f). \end{aligned}$$

If $x(t) \leftrightarrow X(f)$, then

$$x(t - t_0) \leftrightarrow X(f)e^{-j2\pi ft_0}$$

Let $\tau = t - t_0 \Rightarrow t = \tau + t_0, dt = d\tau$.

$$\begin{aligned} \mathcal{F}\{x(t - t_0)\} &= \int_{-\infty}^{\infty} x(t - t_0)e^{-j2\pi ft} dt \\ &= \int_{-\infty}^{\infty} x(\tau)e^{-j2\pi f(\tau + t_0)} d\tau \\ &= e^{-j2\pi ft_0} \int_{-\infty}^{\infty} x(\tau)e^{-j2\pi f\tau} d\tau = e^{-j2\pi ft_0} X(f) \end{aligned}$$

If $x(t) \leftrightarrow X(f)$, then

$$e^{j2\pi f_0 t} x(t) \leftrightarrow X(f - f_0)$$

since,

$$\begin{aligned}\mathcal{F}\{x(t)e^{j2\pi f_0 t}\} &= \int x(t)e^{j2\pi f_0 t}e^{-j2\pi ft}dt \\ &= \int x(t)e^{-j2\pi(f-f_0)t}dt = X(f-f_0)\end{aligned}$$

If $x(t) \leftrightarrow X(f)$ and $a \in \mathbb{R} \setminus \{0\}$, then

$$x(at) \leftrightarrow \frac{1}{|a|}X\left(\frac{f}{a}\right)$$

Let $u = at \Rightarrow t = u/a$, $dt = du/a$.

$$\begin{aligned}\mathcal{F}\{x(at)\} &= \int_{-\infty}^{\infty} x(at)e^{-j2\pi ft}dt \\ &= \int_{-\infty}^{\infty} x(u)e^{-j2\pi fu/a} \cdot \frac{du}{a} = \frac{1}{|a|}X\left(\frac{f}{a}\right)\end{aligned}$$

If $x(t) * h(t) = \int_{-\infty}^{\infty} x(\tau)h(t-\tau)d\tau$, then

$$x(t) * h(t) \leftrightarrow X(f)H(f).$$

Let $y(t) = x(t) * h(t)$.

$$\begin{aligned}\mathcal{F}\{y(t)\} &= \int y(t)e^{-j2\pi ft}dt \\ &= \int \left[\int x(\tau)h(t-\tau)d\tau \right] e^{-j2\pi ft}dt \\ &= \int x(\tau) \left[\int h(t-\tau)e^{-j2\pi ft}dt \right] d\tau \quad (\text{Fubini}) \\ \text{Let } u = t - \tau &\Rightarrow t = u + \tau, dt = du \\ &= \int x(\tau)e^{-j2\pi f\tau} \left[\int h(u)e^{-j2\pi fu}du \right] d\tau \\ &= H(f) \cdot X(f)\end{aligned}$$

□

3 Homework Set 3

3.1 Problem 1

Using Laplace Transforms, find the solution $y(t)$ of the following second-order linear ordinary differential equation given below.

$$\frac{d^2y}{dt^2} + 3\frac{dy}{dt} + 2y = \frac{dx}{dt} + 3x$$

3.1.1 Solution

Notice this problem can be solved with two methods. Denote Method 1 as the Laplace-transform approach and Method 2 as the LTI-operator/Green's function approach.

Proof. Let $x(t), y(t)$ be piecewise-continuous, of exponential order, and assume $x(t) = 0$ for $t < 0$, $y(t) = 0$ for $t < 0$, with $x(0^-) = 0$. Consider

$$\frac{d^2y}{dt^2} + 3\frac{dy}{dt} + 2y = \frac{dx}{dt} + 3x(t).$$

Denote $X(s) = \mathcal{L}\{x(t)\}$ and $Y(s) = \mathcal{L}\{y(t)\}$. With zero initial conditions $y(0) = y'(0) = x(0) = 0$,

$$\mathcal{L}\{y''(t)\} = s^2Y(s), \quad \mathcal{L}\{y'(t)\} = sY(s), \quad \mathcal{L}\{x'(t)\} = sX(s).$$

Taking Laplace transforms of both sides gives

$$s^2Y(s) + 3sY(s) + 2Y(s) = sX(s) + 3X(s),$$

so,

$$Y(s)(s^2 + 3s + 2) = X(s)(s + 3).$$

Thus,

$$Y(s) = H(s)X(s), \quad H(s) = \frac{s + 3}{s^2 + 3s + 2} = \frac{s + 3}{(s + 1)(s + 2)}.$$

We now compute the (causal) impulse response $h(t)$.

$$\frac{s + 3}{(s + 1)(s + 2)} = \frac{A}{s + 1} + \frac{B}{s + 2} \implies s + 3 = A(s + 2) + B(s + 1).$$

Matching coefficients,

$$A + B = 1, \quad 2A + B = 3 \implies A = 2, B = -1.$$

Thus,

$$H(s) = \frac{2}{s+1} - \frac{1}{s+2} \implies h(t) = (2e^{-t} - e^{-2t})u(t),$$

where we have explicitly multiplied by $u(t)$ to enforce causality. By the convolution theorem,

$$y(t) = (h * x)(t) = \int_0^t (2e^{-(t-\tau)} - e^{-2(t-\tau)}) x(\tau) d\tau,$$

which is the desired solution expressed via the causal impulse response. $\square$

Proof. Define the LTI differential operator

$$L[y] := y'' + 3y' + 2y,$$

and consider

$$L[y] = x'(t) + 3x(t).$$

Let $g(t)$ be the causal Green's function satisfying

$$L[g] = \delta(t), \quad g(t) = 0 \text{ for } t < 0.$$

Taking Laplace transforms,

$$G(s) = \mathcal{L}\{g(t)\} = \frac{1}{s^2 + 3s + 2} = \frac{1}{(s+1)(s+2)}.$$

Partial fractions give

$$G(s) = \frac{1}{s+1} - \frac{1}{s+2} \implies g(t) = (e^{-t} - e^{-2t})u(t).$$

Since L is causal and time-invariant, its inverse on causal inputs is convolution with g .

$$y(t) = L^{-1}[x' + 3x] = g * (x' + 3x) = g * x' + 3g * x$$

Using convolution properties for causal signals with $x(0^-) = 0$,

$$(g * x')(t) = (g' * x)(t),$$

so,

$$y(t) = (g' + 3g) * x(t),$$

which shows that the overall impulse response is

$$h(t) = g'(t) + 3g(t).$$

For $t > 0$ we have $g(t) = e^{-t} - e^{-2t}$ and

$$g'(t) = -e^{-t} + 2e^{-2t},$$

so,

$$h(t) = [-e^{-t} + 2e^{-2t} + 3(e^{-t} - e^{-2t})]u(t) = (2e^{-t} - e^{-2t})u(t),$$

where the $u(t)$ factor enforces causality and any δ -terms vanish because $x(0^-) = 0$.

Thus,

$$y(t) = (h * x)(t) = \int_0^t (2e^{-(t-\tau)} - e^{-2(t-\tau)})x(\tau) d\tau,$$

which exactly matches the result from Method 1 with the correct coefficients and explicit causal factor $u(t)$. □

3.2 Problem 2

Find $y(t)$ and $z(t)$ when $x(t) = u(t)$, where $u(t)$ is a step function, by solving the following system of ordinary differential equations using the Laplace Transform given below for all zero initial conditions.

$$\begin{aligned}\frac{dy}{dt} + 2y + z &= \frac{dx}{dt} + x \\ \frac{dz}{dt} + 3z + 2y &= x\end{aligned}$$

3.2.1 Solution

Notice this problem can be solved with two methods. Denote Method 1 as Laplace Transforms and Method 2 as State-Space Convolution by Matrix Exponentials.

Proof. Let $x(t) = u(t)$ be the Heaviside step function, with Laplace transform $\mathcal{L}\{u(t)\} = \frac{1}{s}$. Assume initial conditions $y(0) = z(0) = 0$. Denote $Y(s) = \mathcal{L}\{y(t)\}$, $Z(s) = \mathcal{L}\{z(t)\}$, and $X(s) = \frac{1}{s}$. Then, transform both equations term by term as follows.

$$\begin{aligned}\mathcal{L}\left\{\frac{dy}{dt} + 2y + z\right\} &= \mathcal{L}\left\{\frac{dx}{dt} + x\right\} \\ \Rightarrow (sY(s) + 2Y(s) + Z(s)) &= sX(s) + X(s) \\ \Rightarrow (s+2)Y(s) + Z(s) &= \frac{s+1}{s}\end{aligned}\tag{1}$$

$$\begin{aligned}\mathcal{L}\left\{\frac{dz}{dt} + 3z + 2y\right\} &= \mathcal{L}\{x(t)\} \\ \Rightarrow (sZ(s) + 3Z(s) + 2Y(s)) &= X(s) \\ \Rightarrow (s+3)Z(s) + 2Y(s) &= \frac{1}{s}\end{aligned}\tag{2}$$

Solve (1) for $Z(s)$:

$$Z(s) = \frac{s+1}{s} - (s+2)Y(s)\tag{3}$$

Substitute (3) into (2):

$$(s+3)\left[\frac{s+1}{s} - (s+2)Y(s)\right] + 2Y(s) = \frac{1}{s}$$

Distribute:

$$(s+3)\frac{s+1}{s} - (s+3)(s+2)Y(s) + 2Y(s) = \frac{1}{s}$$

Group terms:

$$[-(s+3)(s+2)+2]Y(s) = \frac{1}{s} - (s+3)\frac{s+1}{s}$$

Compute the left side:

$$-(s^2 + 5s + 6) + 2 = -s^2 - 5s - 4$$

Compute the right side:

$$\frac{1 - (s+3)(s+1)}{s} = \frac{1 - (s^2 + 4s + 3)}{s} = \frac{-s^2 - 4s - 2}{s}$$

So,

$$(-s^2 - 5s - 4)Y(s) = \frac{-s^2 - 4s - 2}{s} \Rightarrow Y(s) = \frac{s^2 + 4s + 2}{s(s^2 + 5s + 4)}$$

Now compute $Z(s)$ using (3):

$$Z(s) = \frac{s+1}{s} - (s+2)Y(s) = \frac{s+1}{s} - (s+2) \cdot \frac{s^2 + 4s + 2}{s(s^2 + 5s + 4)}$$

Leave this in rational form. Now, apply the inverse Laplace transform using partial fractions and standard table lookups. Then, denote the following.

$$H_y(s) = \frac{s^2 + 4s + 2}{s(s^2 + 5s + 4)} = \frac{s^2 + 4s + 2}{s(s+1)(s+4)}$$

Since the denominator factors as:

$$s(s+1)(s+4)$$

we can write:

$$H_y(s) = \frac{A}{s} + \frac{B}{s+1} + \frac{C}{s+4}$$

Multiply both sides by the denominator:

$$s^2 + 4s + 2 = A(s+1)(s+4) + Bs(s+4) + Cs(s+1)$$

Solving for the constants:

Case 1: Let $s = 0$:

$$0 + 0 + 2 = A(1)(4) \Rightarrow A = \frac{1}{2}$$

Case 2: Let $s = -1$:

$$1 - 4 + 2 = B(-1)(3) \Rightarrow -1 = -3B \Rightarrow B = \frac{1}{3}$$

Case 3: Let $s = -4$:

$$16 - 16 + 2 = C(-4)(-3) \Rightarrow 2 = 12C \Rightarrow C = \frac{1}{6}$$

Therefore:

$$H_y(s) = \frac{1}{2s} + \frac{1}{3(s+1)} + \frac{1}{6(s+4)}$$

Now apply inverse Laplace transform using standard Laplace pairs:

$$\mathcal{L}^{-1}\left\{\frac{1}{s}\right\} = 1, \quad \mathcal{L}^{-1}\left\{\frac{1}{s+a}\right\} = e^{-at}$$

Thus:

$$Y(s) = \frac{1}{2s} + \frac{1}{3(s+1)} + \frac{1}{6(s+4)} \Rightarrow y(t) = \left(\frac{1}{2} + \frac{1}{3}e^{-t} + \frac{1}{6}e^{-4t}\right), \quad t \geq 0$$

This function is continuous and differentiable on $\mathbb{R}_{\geq 0}$, and satisfies the Laplace-inverted form of $Y(s)$. It is clear that the exponential terms correspond to the distinct poles of the system, and the constants match the residues at each pole. Hence, this time-domain representation is unique by Lerch's theorem and matches the expected causal LTI solution. Thus, the time-domain solution is as follows.

$$y(t) = \left(\frac{1}{2} + \frac{1}{3}e^{-t} + \frac{1}{6}e^{-4t}\right)u(t)$$

Now compute $z(t)$ by inverse-transforming $Z(s)$ as follows.

$$Z(s) = \frac{s+1}{s} - (s+2)Y(s) = \frac{s+1}{s} - (s+2)\left(\frac{1}{2s} + \frac{1}{3(s+1)} + \frac{1}{6(s+4)}\right)$$

Distribute $(s+2)$ across the partial fractions. Denote the overall numerator by ξ , where $\xi = \xi_1 - \xi_2$, and:

$$\xi_1 = (s+1), \quad \xi_2 = (s+2)\left(\frac{1}{2} + \frac{1}{3} \cdot \frac{s}{s+1} + \frac{1}{6} \cdot \frac{s}{s+4}\right)$$

Compute each term explicitly.

First, expand:

$$\xi_2 = \frac{s+2}{2s} + \frac{s+2}{3(s+1)} + \frac{s+2}{6(s+4)}$$

Then write:

$$Z(s) = \xi = \left(\frac{s+1}{s} \right) - \left[\frac{s+2}{2s} + \frac{s+2}{3(s+1)} + \frac{s+2}{6(s+4)} \right]$$

Now simplify $\xi_1 - \xi_2$ term by term.

$$\begin{aligned} \frac{s+1}{s} - \frac{s+2}{2s} &= \frac{2(s+1) - (s+2)}{2s} = \frac{s}{2s} = \frac{1}{2} \\ \Rightarrow \frac{s+2}{3(s+1)} &= \frac{1}{3} + \frac{1}{3(s+1)}, \quad \frac{s+2}{6(s+4)} = \frac{1}{6} + \frac{-1}{3(s+4)} \end{aligned}$$

Combine all:

$$Z(s) = \frac{1}{2} - \left(\frac{1}{3} + \frac{1}{3(s+1)} + \frac{1}{6} - \frac{1}{3(s+4)} \right)$$

Simplify constants:

$$Z(s) = \left(\frac{1}{2} - \frac{1}{3} - \frac{1}{6} \right) - \left(\frac{1}{3(s+1)} - \frac{1}{3(s+4)} \right) = 0 - \left(\frac{1}{3(s+1)} - \frac{1}{3(s+4)} \right)$$

Thus,

$$Z(s) = -\frac{1}{3(s+1)} + \frac{1}{3(s+4)}.$$

Apply inverse Laplace transforms:

$$z(t) = -\frac{1}{3}e^{-t} + \frac{1}{3}e^{-4t}, \quad t \geq 0$$

Therefore, the time-domain solutions are continuous and differentiable on $\mathbb{R}_{\geq 0}$, and satisfy the original coupled system under zero initial conditions.

$$\begin{aligned} y(t) &= \left(\frac{1}{2} + \frac{1}{3}e^{-t} + \frac{1}{6}e^{-4t} \right) u(t) \\ z(t) &= \frac{1}{3} (e^{-4t} - e^{-t}) \end{aligned}$$

□

Proof. Rewriting the coupled system

$$\begin{cases} \frac{dy}{dt} + 2y + z = \frac{dx}{dt} + x, \\ \frac{dz}{dt} + 3z + 2y = x \end{cases}$$

as a first-order linear vector ODE of the form

$$\frac{d}{dt} \begin{bmatrix} y(t) \\ z(t) \end{bmatrix} = \underbrace{\begin{bmatrix} -2 & -1 \\ -2 & -3 \end{bmatrix}}_A \begin{bmatrix} y(t) \\ z(t) \end{bmatrix} + \underbrace{\begin{bmatrix} 1 \\ 1 \end{bmatrix}}_b x(t) + \underbrace{\begin{bmatrix} 1 \\ 0 \end{bmatrix}}_c \frac{dx}{dt}$$

This is a linear time-invariant (LTI) nonhomogeneous system with both input and input derivative. By the variation of parameters theorem for matrix systems (see Coddington and Levinson, *Theory of ODEs*), the solution with zero initial condition is

$$\begin{bmatrix} y(t) \\ z(t) \end{bmatrix} = \int_0^t e^{A(t-\tau)} \left(b x(\tau) + c \frac{dx}{d\tau} \right) d\tau$$

This form is particularly effective for analyzing systems driven by combinations of step and impulse inputs. To compute e^{At} , we proceed by Laplace methods. Recall that the Laplace transform of the matrix exponential is the resolvent, with the general form given below.

$$\mathcal{L}\{e^{At}\} = (sI - A)^{-1}$$

Given

$$A = \begin{bmatrix} -2 & -1 \\ -2 & -3 \end{bmatrix} \Rightarrow sI - A = \begin{bmatrix} s+2 & 1 \\ 2 & s+3 \end{bmatrix}$$

We compute the inverse:

$$(sI - A)^{-1} = \frac{1}{\det(sI - A)} \begin{bmatrix} s+3 & -1 \\ -2 & s+2 \end{bmatrix} = \frac{1}{s^2 + 5s + 4} \begin{bmatrix} s+3 & -1 \\ -2 & s+2 \end{bmatrix}$$

This matrix is the Laplace-domain resolvent $R(s)$. The inverse Laplace transform gives the matrix exponential:

$$e^{At} = \mathcal{L}^{-1}\{R(s)\}$$

Since the characteristic polynomial factors as $(s+1)(s+4)$, the entries of e^{At} are linear combinations of e^{-t} and e^{-4t} . That is,

$$e^{At} = \alpha_1 e^{-t} + \alpha_2 e^{-4t}, \quad \text{entry-wise}$$

Returning to the solution:

$$\begin{bmatrix} y(t) \\ z(t) \end{bmatrix} = e^{At} c + \int_0^t e^{A(t-\tau)} b x(\tau) d\tau$$

Given that $x(t) = u(t)$, substitute $\frac{dx}{dt} = \delta(t)$. The convolution integral simplifies to

$$\int_0^t e^{A(t-\tau)} c \delta(\tau) d\tau = e^{At} c,$$

and

$$\int_0^t e^{A(t-\tau)} b x(\tau) d\tau = \int_0^t e^{A(t-\tau)} b d\tau.$$

So,

$$\begin{bmatrix} y(t) \\ z(t) \end{bmatrix} = e^{At} c + \left(\int_0^t e^{A\tau} d\tau \right) b$$

Let us denote $H(t) = e^{At}$. Then the output $y(t)$, the first component of the state, can be expressed as:

$$y(t) = \int_0^t h_1(t-\tau) x(\tau) d\tau + \int_0^t h_2(t-\tau) \frac{dx}{d\tau} d\tau$$

where h_1 and h_2 are the first-row entries of the matrix kernels $H(t)b$ and $H(t)c$, respectively. These represent the impulse responses of the system with respect to $x(t)$ and $\frac{dx}{dt}$ respectively, and provides an analytically exact solution framework. $\square$

3.3 Problem 3

Using the Z-Transform, when $x(n)$ is a discrete step function $u(n)$ (i.e. 0 wherever $n < 0$ and $n \geq 0$), find the casual discrete-time signal $y(n)$ defined by the following linear difference equation.

$$y(n) + \frac{5}{6}y(n-1) + \frac{1}{6}y(n-2) = x(n) + \frac{1}{2}x(n-1)$$

3.3.1 Solution

Notice this problem can be solved with two methods. Denote Method 1 as Z-Transform and Method 2 as Direct Recurrence.

Proof. Given the linear constant-coefficient difference equation:

$$y(n) + \frac{5}{6}y(n-1) + \frac{1}{6}y(n-2) = x(n) + \frac{1}{2}x(n-1)$$

with $x(n) = u(n)$, the discrete-time unit step function. We assume $y(n) = 0$ for $n < 0$ (causal solution), and all sequences are defined for $n \in \mathbb{N}_0$.

Let $Y(z) = \mathcal{Z}\{y(n)\}$, $X(z) = \mathcal{Z}\{x(n)\}$, where:

$$X(z) = \mathcal{Z}\{u(n)\} = \frac{1}{1 - z^{-1}}, \quad |z| > 1$$

Apply the Z-transform to both sides using linearity and the time-shift property $\mathcal{Z}\{y(n-k)\} = z^{-k}Y(z)$ under zero initial conditions. Then, the left hand side (LHS) is

$$Y(z) + \frac{5}{6}z^{-1}Y(z) + \frac{1}{6}z^{-2}Y(z) = Y(z) \left(1 + \frac{5}{6}z^{-1} + \frac{1}{6}z^{-2}\right),$$

and the right hand side (RHS) is

$$X(z) + \frac{1}{2}z^{-1}X(z) = X(z) \left(1 + \frac{1}{2}z^{-1}\right).$$

Thus,

$$Y(z) \left(1 + \frac{5}{6}z^{-1} + \frac{1}{6}z^{-2}\right) = \frac{1 + \frac{1}{2}z^{-1}}{1 - z^{-1}}, \quad |z| > 1$$

Multiply both sides by z^2 to obtain the following polynomial ratio.

$$Y(z) \left(z^2 + \frac{5}{6}z + \frac{1}{6}\right) = \frac{z^2 + \frac{1}{2}z}{z - 1}$$

Simplifying the numerator as follows.

$$z^2 + \frac{1}{2}z = z \left(z + \frac{1}{2} \right)$$

Hence,

$$Y(z) = \frac{z(z + \frac{1}{2})}{(z - 1)(z^2 + \frac{5}{6}z + \frac{1}{6})}$$

Now, factoring the denominator.

$$z^2 + \frac{5}{6}z + \frac{1}{6} = \left(z + \frac{1}{2} \right) \left(z + \frac{1}{3} \right)$$

Thus,

$$Y(z) = \frac{z(z + \frac{1}{2})}{(z - 1)(z + \frac{1}{2})(z + \frac{1}{3})} = \frac{z}{(z - 1)(z + \frac{1}{3})}$$

Now apply partial fraction decomposition.

$$\frac{z}{(z - 1)(z + \frac{1}{3})} = \frac{A}{z - 1} + \frac{B}{z + \frac{1}{3}}$$

Multiply both sides by the denominator as follows.

$$z = A \left(z + \frac{1}{3} \right) + B(z - 1)$$

Solve for A, B as follows.

Case 1: Let $z = 1$: $1 = A \left(1 + \frac{1}{3} \right) \Rightarrow A = \frac{3}{4}$

Case 2: Let $z = -\frac{1}{3}$: $-\frac{1}{3} = B \left(-\frac{1}{3} - 1 \right) = B \left(-\frac{4}{3} \right) \Rightarrow B = \frac{1}{4}$

So,

$$Y(z) = \frac{3}{4} \cdot \frac{1}{z - 1} + \frac{1}{4} \cdot \frac{1}{z + \frac{1}{3}}$$

Now take the inverse Z-transform using standard tables:

$$\mathcal{Z}^{-1} \left\{ \frac{1}{z - a} \right\} = a^n u(n)$$

Simplifying the following.

$$y(n) = \frac{3}{4}(1)^n + \frac{1}{4} \left(-\frac{1}{3} \right)^n, \quad n \geq 0$$

Therefore, the function is causal, bounded, and satisfies the initial value $y(0) = 1$.

$$y(n) = \left(\frac{3}{4} + \frac{1}{4} \left(-\frac{1}{3} \right)^n \right) u(n), \quad n \in \mathbb{N}_0$$

□

Proof. Consider the causal discrete-time linear difference equation

$$y(n) + \frac{5}{6}y(n-1) + \frac{1}{6}y(n-2) = x(n) + \frac{1}{2}x(n-1), \quad n \geq 0,$$

where $x(n) = u(n)$ denotes the unit step function. Assume zero initial conditions:

$$y(-1) = y(-2) = 0$$

By the standard theory of linear constant-coefficient difference equations (see Thm. 2.1 in Elaydi's *An Introduction to Difference Equations*), the general solution is the sum of a homogeneous and particular component:

$$y(n) = y_h(n) + y_p(n)$$

where $y_h(n)$ solves the homogeneous equation and $y_p(n)$ any particular solution.

Consider the following homogeneous difference equation.

$$y_h(n) + \frac{5}{6}y_h(n-1) + \frac{1}{6}y_h(n-2) = 0$$

The associated characteristic polynomial is:

$$r^2 + \frac{5}{6}r + \frac{1}{6} = 0$$

Multiply by 6 to clear denominators:

$$6r^2 + 5r + 1 = 0$$

By the quadratic formula:

$$r = \frac{-5 \pm \sqrt{25 - 24}}{12} = \frac{-5 \pm 1}{12}$$

Hence,

$$r_1 = -\frac{1}{3}, \quad r_2 = -\frac{1}{2}$$

Thus, the homogeneous solution is as follows.

$$y_h(n) = C_1 \left(-\frac{1}{3} \right)^n + C_2 \left(-\frac{1}{2} \right)^n, \quad n \geq 0$$

Since $x(n) = u(n)$ is constant for $n \geq 0$, seek a constant particular solution $y_p(n) = K$.

Substitute $y(n) = y(n-1) = y(n-2) = K$ into the original recurrence:

$$K + \frac{5}{6}K + \frac{1}{6}K = 1 + \frac{1}{2}$$

Simplifying as follows.

$$K \left(1 + \frac{5}{6} + \frac{1}{6} \right) = \frac{3}{2} \Rightarrow 2K = \frac{3}{2} \Rightarrow K = \frac{3}{4}$$

It follows that

$$y_p(n) = \frac{3}{4}.$$

Thus, the general solution is

$$y(n) = \frac{3}{4} + C_1 \left(-\frac{1}{3} \right)^n + C_2 \left(-\frac{1}{2} \right)^n.$$

Now, use the zero initial conditions. For $x(n) = u(n)$:

$$x(0) = 1, \quad x(-1) = 0$$

Case 1: At $n = 0$:

$$y(0) + \frac{5}{6}y(-1) + \frac{1}{6}y(-2) = x(0) + \frac{1}{2}x(-1) \Rightarrow y(0) = 1$$

Case 2: At $n = 1$:

$$y(1) + \frac{5}{6}y(0) + \frac{1}{6}y(-1) = x(1) + \frac{1}{2}x(0) \Rightarrow y(1) + \frac{5}{6}(1) = 1 + \frac{1}{2} \Rightarrow y(1) = \frac{2}{3}$$

Thus,

$$\begin{cases} y(0) = \frac{3}{4} + C_1 + C_2 = 1, \\ y(1) = \frac{3}{4} - \frac{1}{3}C_1 - \frac{1}{2}C_2 = \frac{2}{3}. \end{cases}$$

Solving:

$$\begin{aligned} C_1 + C_2 &= \frac{1}{4}, \\ -\frac{1}{3}C_1 - \frac{1}{2}C_2 &= -\frac{1}{12}. \end{aligned}$$

Multiply the second equation by 6:

$$-2C_1 - 3C_2 = -\frac{1}{2}$$

Multiply the first by 2:

$$2C_1 + 2C_2 = \frac{1}{2}.$$

Adding both as follows.

$$(-2C_1 - 3C_2) + (2C_1 + 2C_2) = 0C_1 - C_2 = 0 \Rightarrow C_2 = 0.$$

The, $C_1 = \frac{1}{4}$.

Hence,

$$C_1 = \frac{1}{4}, \quad C_2 = 0.$$

Therefore, as $n \rightarrow \infty$, the transient term $\left(-\frac{1}{3}\right)^n \rightarrow 0$, it is clear that

$$\lim_{n \rightarrow \infty} y(n) = \frac{3}{4},$$

, so that both Method 1 and Method 2 satisfy the existence and uniqueness theorem, and the following solution is obtained.

$$y(n) = \frac{3}{4} + \frac{1}{4} \left(-\frac{1}{3}\right)^n, \quad n \geq 0$$

□

4 Homework Set 4

4.1 Problem 1

Find the Transfer Function of the LTI system described by the differential equation

$$\frac{d^2y(t)}{dt^2} + 3\frac{dy(t)}{dt} + 2y(t) = \frac{dx(t)}{dt} - x(t)$$

where $x(t)$ and $y(t)$ denote the input and output of the system, respectively.

1. Find the Impulse Response Function and the Gain & Phase Functions of this system.
2. Determine the bandwidth of this system using the half-max (-3 dB) convention.
3. If the input signal $x(t)$ is sinusoidal with frequencies that can range from near-zero to 10 rad/s, determine the minimum sampling rate required for the continuous-time signals $x(t)$ and $y(t)$ to avoid aliasing for the bandwidth determined above.
4. Determine the sampling rate required to avoid aliasing for the highest input frequency of 10 rad/s.

4.1.1 Solution

1.

Theorem 4.1. *The transfer function $H(s) = \frac{Y(s)}{X(s)}$ of the given system is:*

$$H(s) = \frac{s-1}{(s+1)(s+2)}.$$

Proof. Taking the Laplace transform of both sides (assuming zero initial conditions):

$$\mathcal{L}\{y''(t) + 3y'(t) + 2y(t)\} = \mathcal{L}\{x'(t) - x(t)\},$$

$$s^2Y(s) + 3sY(s) + 2Y(s) = sX(s) - X(s),$$

$$Y(s)(s^2 + 3s + 2) = X(s)(s - 1),$$

$$\therefore H(s) = \frac{Y(s)}{X(s)} = \frac{s-1}{s^2 + 3s + 2} = \frac{s-1}{(s+1)(s+2)}.$$

□

2.

Theorem 4.2. *The impulse response of the system is:*

$$h(t) = (-2e^{-t} + 3e^{-2t})u(t).$$

Proof. Perform a partial fraction decomposition:

$$\frac{s-1}{(s+1)(s+2)} = \frac{A}{s+1} + \frac{B}{s+2}.$$

Solving for A and B :

$$s-1 = A(s+2) + B(s+1) \Rightarrow (A+B)s + (2A+B) = s-1.$$

Matching coefficients:

$$A+B=1, \quad 2A+B=-1 \quad \Rightarrow \quad A=-2, \quad B=3.$$

Thus:

$$H(s) = \frac{-2}{s+1} + \frac{3}{s+2} \Rightarrow h(t) = (-2e^{-t} + 3e^{-2t})u(t), \quad t \geq 0.$$

□

3.

Theorem 4.3. *The magnitude and phase of the frequency response $H(j\omega)$ are:*

$$|H(j\omega)| = \frac{1}{\sqrt{\omega^2 + 4}}, \quad \angle H(j\omega) = (\pi - \tan^{-1} \omega) - \tan^{-1} \omega - \tan^{-1} \left(\frac{\omega}{2} \right),$$

with phase unwrapped for continuity.

Proof. Let

$$H(j\omega) = \frac{j\omega - 1}{(j\omega + 1)(j\omega + 2)}.$$

Then:

$$|H(j\omega)| = \frac{\sqrt{\omega^2 + 1}}{\sqrt{(\omega^2 + 1)(\omega^2 + 4)}} = \frac{1}{\sqrt{\omega^2 + 4}}.$$

Using $\arg(ab) = \arg a + \arg b$, the total phase is:

$$\angle H(j\omega) = \arg(j\omega - 1) - \arg(j\omega + 1) - \arg(j\omega + 2).$$

Since $\arg(j\omega - 1) = \pi - \tan^{-1} \omega$ for $\omega > 0$, we obtain:

$$\angle H(j\omega) = (\pi - \tan^{-1} \omega) - \tan^{-1} \omega - \tan^{-1} \left(\frac{\omega}{2} \right).$$

□

By Nyquist, the minimum sampling angular frequency to avoid aliasing is

$$\Omega_s > 2\omega_B = 4 \text{ rad/s}, \quad \text{equivalently } f_s > \frac{4}{2\pi} \approx 0.637 \text{ Hz}.$$

4.

Lemma 4.4. *The -3 dB bandwidth of the system is:*

$$\omega_B = 2 \text{ rad/s.}$$

Proof. The bandwidth is defined where the magnitude drops by $1/\sqrt{2}$ from its maximum:

$$|H(j\omega_B)| = \frac{|H(0)|}{\sqrt{2}}.$$

Since $|H(0)| = \frac{1}{2}$,

$$\frac{1}{\sqrt{\omega_B^2 + 4}} = \frac{1}{2\sqrt{2}} \Rightarrow \omega_B^2 + 4 = 8 \Rightarrow \omega_B = 2 \text{ rad/s.}$$

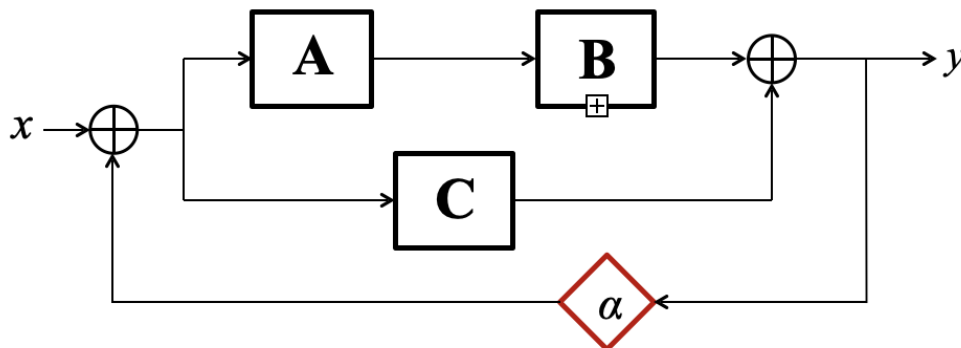
□

For inputs up to $\omega_{\max} = 10 \text{ rad/s}$,

$$\Omega_s > 2\omega_{\max} = 20 \text{ rad/s,} \quad \text{equivalently } f_s > \frac{20}{2\pi} \approx 3.183 \text{ Hz.}$$

4.2 Problem 2

Find the Transfer Function from input $x(t)$ to output $y(t)$ in terms of the Transfer Functions $A(\omega)$, $B(\omega)$, $C(\omega)$ of the system shown below for the three LTI subsystem components and the scalar feedback α .



4.2.1 Solution

Consider the closed-loop system in Fig. ??, and let the signal after the left summing junction be

$$e(t) = x(t) - \alpha y(t).$$

This signal $e(t)$ is fed in parallel into the two LTI subsystems A and C , whose outputs are added and then passed through B before the right summing junction. In the frequency domain, let $E(\omega)$ and $Y(\omega)$ denote the Fourier transforms of $e(t)$ and $y(t)$, respectively. Then,

$$\text{output of } A : A(\omega)E(\omega), \quad \text{output of } C : C(\omega)E(\omega),$$

so the summed signal at the input of B is

$$V(\omega) = A(\omega)E(\omega) + C(\omega)E(\omega) = (A(\omega) + C(\omega))E(\omega).$$

Passing this through B gives

$$Y(\omega) = B(\omega)V(\omega) = B(\omega)(A(\omega) + C(\omega))E(\omega).$$

Thus, the effective forward-path transfer function from e to y is

$$G(\omega) = B(\omega)(A(\omega) + C(\omega)).$$

The feedback path contains only the scalar gain α , so its transfer function is

$$H_f(\omega) = \alpha.$$

Theorem 4.5. *The closed-loop transfer function from input $x(t)$ to output $y(t)$ is*

$$H(\omega) = \frac{Y(\omega)}{X(\omega)} = \frac{B(\omega)(A(\omega) + C(\omega))}{1 + \alpha B(\omega)(A(\omega) + C(\omega))}.$$

Proof. From the feedback relation, $e(t) = x(t) - \alpha y(t)$, we have in the frequency domain

$$E(\omega) = X(\omega) - \alpha Y(\omega).$$

Using the forward-path relation $Y(\omega) = G(\omega)E(\omega)$ with $G(\omega) = B(\omega)(A(\omega) + C(\omega))$, we obtain

$$Y(\omega) = G(\omega)(X(\omega) - \alpha Y(\omega)) = G(\omega)X(\omega) - \alpha G(\omega)Y(\omega).$$

Rearranging terms,

$$[1 + \alpha G(\omega)]Y(\omega) = G(\omega)X(\omega),$$

so

$$H(\omega) = \frac{Y(\omega)}{X(\omega)} = \frac{G(\omega)}{1 + \alpha G(\omega)} = \frac{B(\omega)(A(\omega) + C(\omega))}{1 + \alpha B(\omega)(A(\omega) + C(\omega))}.$$

This matches the block-diagram structure in which A and C form a parallel forward path into B , with the scalar gain α in the feedback loop. $\square$

5 Homework Set 5

5.1 Problem 1

Using Matlab, generate the discrete-time signal

$$x(n) = \cos\left(\frac{2\pi n}{250}\right) + \sin\left(\frac{2\pi n}{100} + \frac{\pi}{n}\right) + cw(n)$$

for $n = 1, \dots, 1000$ for $c = 1$, where $w(n)$ denotes a sequence of independent normal random samples (white noise) that are generated in Matlab by use of the command `randn`. Compute the Discrete Fourier Transform of $x(n)$ and plot its Real and Imaginary parts, as well as its Magnitude and Phase. Comment on whether you can visually discern more easily the presence of periodicities within the time-series data $x(n)$ in the time-domain or in the frequency-domain. Compute the autocorrelation function of $x(n)$ and comment on the ability to discern in it (visually) the periodicities present within $x(n)$. When you change the value of c , ensure to find the maximum value for which you can still discern visually the presence of periodicities within the time-series data $x(n)$ in the time-domain or in the frequency-domain.

5.1.1 Solution

Proof. Given

$$x[n] = \cos\left(\frac{2\pi n}{250}\right) + \sin\left(\frac{2\pi n}{100} + \frac{\pi}{4}\right) + cw[n], \quad n = 1, \dots, 1000,$$

where $w[n]$ is i.i.d., zero-mean, unit-variance white noise, and $c > 0$.

Figure 5 shows a realization of $x[n]$ for $c = 1$.

Write

$$x[n] = s[n] + cw[n],$$

with deterministic part

$$s[n] := \cos\left(\frac{2\pi n}{250}\right) + \sin\left(\frac{2\pi n}{100} + \frac{\pi}{4}\right).$$

Each sinusoid has amplitude 1, so using $\mathbb{E}\{\cos^2(\cdot)\} = \mathbb{E}\{\sin^2(\cdot)\} = \frac{1}{2}$ and cross terms averaging to 0 over many periods, the average power of $s[n]$ is

$$P_s = \mathbb{E}\{s^2[n]\} \approx \frac{1}{2} + \frac{1}{2} = 1.$$

The noise term $cw[n]$ has variance c^2 , so its power is $P_w = c^2$. Thus, the signal-to-noise ratio is

$$\text{SNR} = \frac{P_s}{P_w} = \frac{1}{c^2}.$$

For $c = 1$, $\text{SNR} = 1$ or 0 dB, so the time-domain trace is expected to be strongly noise-contaminated, which is visible in Figure 5: the oscillations are present but difficult to discern purely by eye.

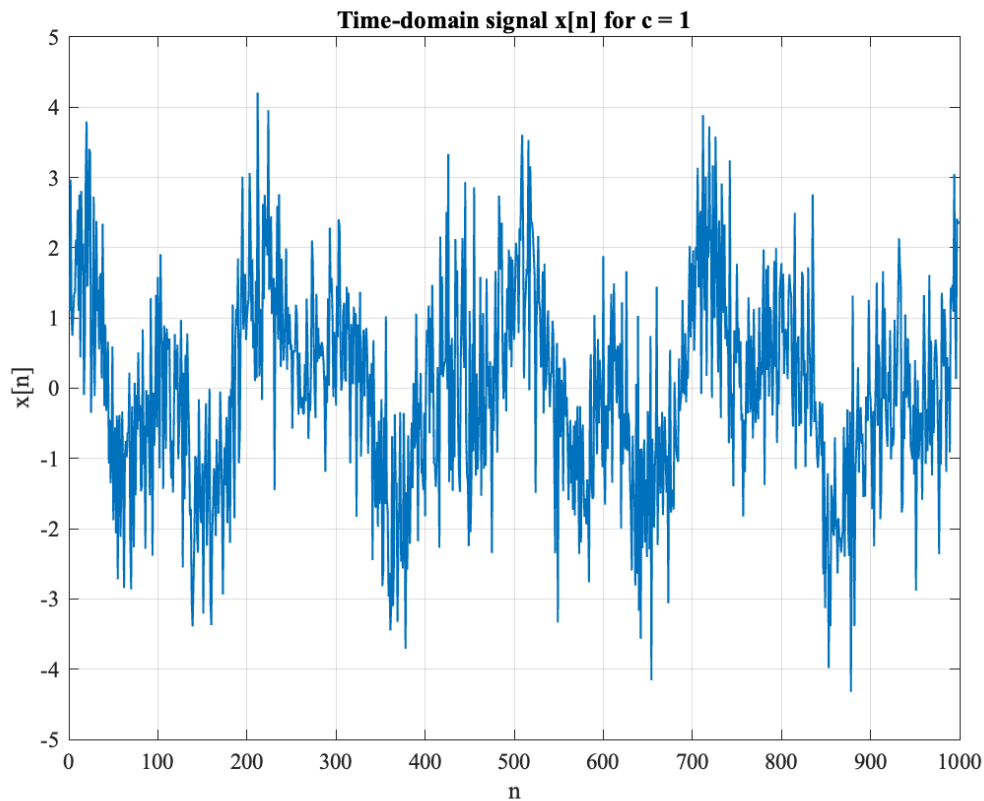


Figure 5: Time-domain signal $x[n]$ for $c = 1$.

Recall the N -point DFT is

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j2\pi kn/N}, \quad k = 0, \dots, N-1.$$

For the cosine term,

$$\cos\left(\frac{2\pi n}{250}\right) = \frac{1}{2}e^{j2\pi n/250} + \frac{1}{2}e^{-j2\pi n/250}.$$

For the sine term,

$$\sin\left(\frac{2\pi n}{100} + \frac{\pi}{4}\right) = \frac{1}{2j} \left(e^{j(\frac{2\pi n}{100} + \frac{\pi}{4})} - e^{-j(\frac{2\pi n}{100} + \frac{\pi}{4})} \right).$$

Thus, $s[n]$ is a sum of four complex exponentials of the form $A_\ell e^{j\omega_\ell n}$ with

$$\omega_1 = \frac{2\pi}{250}, \quad \omega_2 = -\frac{2\pi}{250}, \quad \omega_3 = \frac{2\pi}{100}, \quad \omega_4 = -\frac{2\pi}{100},$$

and corresponding complex amplitudes A_ℓ obtained from the above decompositions.

For an exponential $x[n] = e^{j2\pi k_0 n/N}$ (no noise) with integer k_0 ,

$$X[k] = \sum_{n=0}^{N-1} e^{j2\pi k_0 n/N} e^{-j2\pi kn/N} = \sum_{n=0}^{N-1} e^{j2\pi(k_0-k)n/N}.$$

If $k = k_0$, the exponent is 0 and

$$X[k_0] = \sum_{n=0}^{N-1} 1 = N.$$

If $k \neq k_0$, we have a finite geometric series:

$$X[k] = \frac{1 - e^{j2\pi(k_0-k)}}{1 - e^{j2\pi(k_0-k)/N}} = 0,$$

since $e^{j2\pi(k_0-k)} = 1$. Thus, the DFT is a single spectral line at $k = k_0$.

The first sinusoid has normalized frequency $\omega_1 = 2\pi/250$, so its cycles-per-sample frequency is $f_1 = \omega_1/(2\pi) = 1/250$. The bin index is

$$k_1 = f_1 N = \frac{1}{250} 1000 = 4.$$

Similarly, for the second sinusoid, $f_2 = 1/100$ and

$$k_2 = f_2 N = \frac{1}{100} 1000 = 10.$$

The negative-frequency components appear at $N - k_1 = 996$ and $N - k_2 = 990$. Each positive/negative frequency pair contributes two lines of equal magnitude and conjugate-symmetric phase.

Thus, for the noiseless signal $s[n]$, the DFT $S[k]$ has nonzero entries only at $k \in \{4, 10, 990, 996\}$, with magnitudes proportional to $N/2$ and phases determined by the sinusoid phases.

The noise term $c w[n]$ has DFT

$$W[k] = \sum_{n=0}^{N-1} c w[n] e^{-j2\pi kn/N}.$$

Because $w[n]$ is white, zero-mean, unit-variance, the real and imaginary parts of $W[k]$ are approximately i.i.d. Gaussian with variance proportional to $c^2 N$; hence $W[k]$ contributes a *flat spectral floor* (same expected power for all k).

Thus, the full DFT

$$X[k] = S[k] + W[k]$$

shows sharp sinusoidal lines at $k = 4, 10, 990, 996$ on top of a broadband noise floor, exactly as observed in Figure 6.

Figure 7 shows time-domain traces for $c \in \{0.25, 0.5, 1, 2, 4\}$.

From the SNR,

$$\text{SNR} = \frac{P_s}{P_w} = \frac{1}{c^2}, \quad \text{SNR}_{\text{dB}} = -20 \log_{10} c.$$

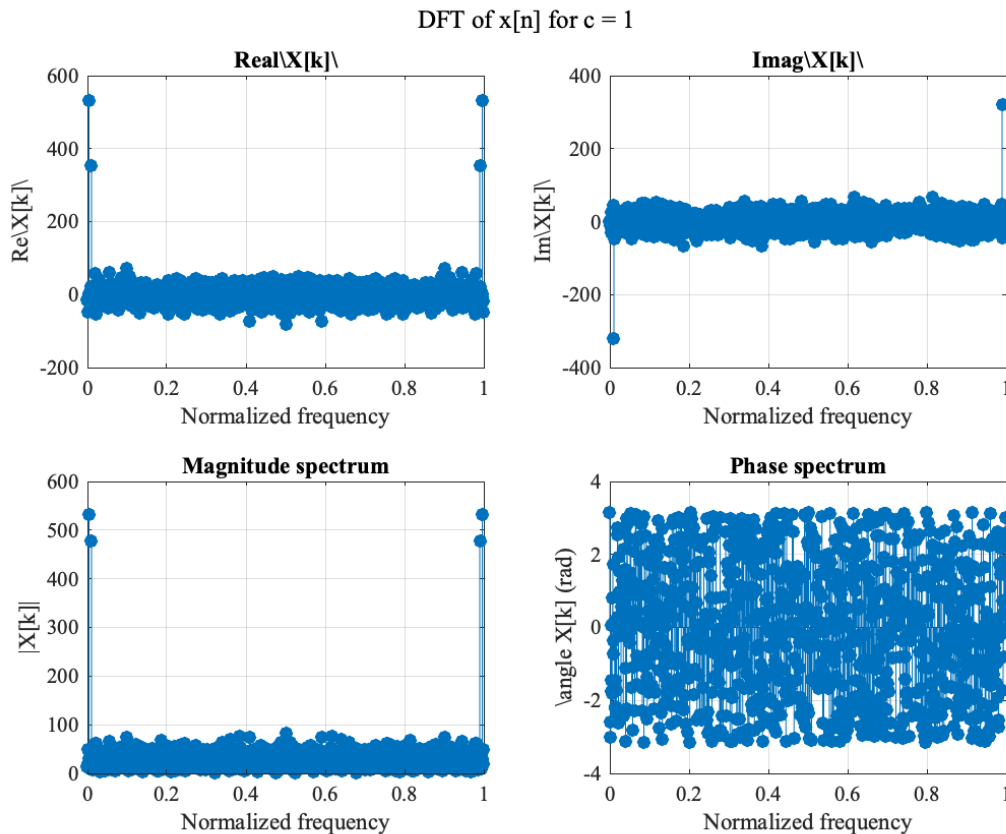


Figure 6: DFT components of $x[n]$ for $c = 1$.

A reasonable criterion for time-domain visibility is $\text{SNR}_{\text{dB}} \gtrsim 0$ dB, i.e. $P_s \gtrsim P_w$, which yields approximately

$$c \lesssim 1.$$

Indeed, in Figure 7 the sinusoidal structure is visually clear for $c = 0.25$ and $c = 0.5$, marginal at $c = 1$, and essentially lost for $c = 2$ and $c = 4$.

In the frequency domain, consider a sinusoid line magnitude of order $N/2$ and a noise magnitude per bin on the order of $c\sqrt{N}$; then a per-bin spectral SNR is approximately

$$\text{SNR}_{\text{bin}} \approx \frac{(N/2)^2}{c^2 N} = \frac{N}{4c^2}.$$

For $N = 1000$ and $c = 4$ this gives $\text{SNR}_{\text{bin}} \approx 1000/(4 \cdot 16) \approx 15.6$, or about 12 dB, which agrees with the observation that the spectral lines remain clearly visible in Figure 8 even for $c = 4$.

Thus, a reasonable quantitative summary is:

$$c_{\text{max,time}} \approx 1, \quad c_{\text{max,freq}} \gg 1 \text{ (at least up to } c = 4 \text{ in our plots).}$$

Figure 9 shows the biased autocorrelation estimate $\hat{R}_{xx}[m]$ for $c = 1$.

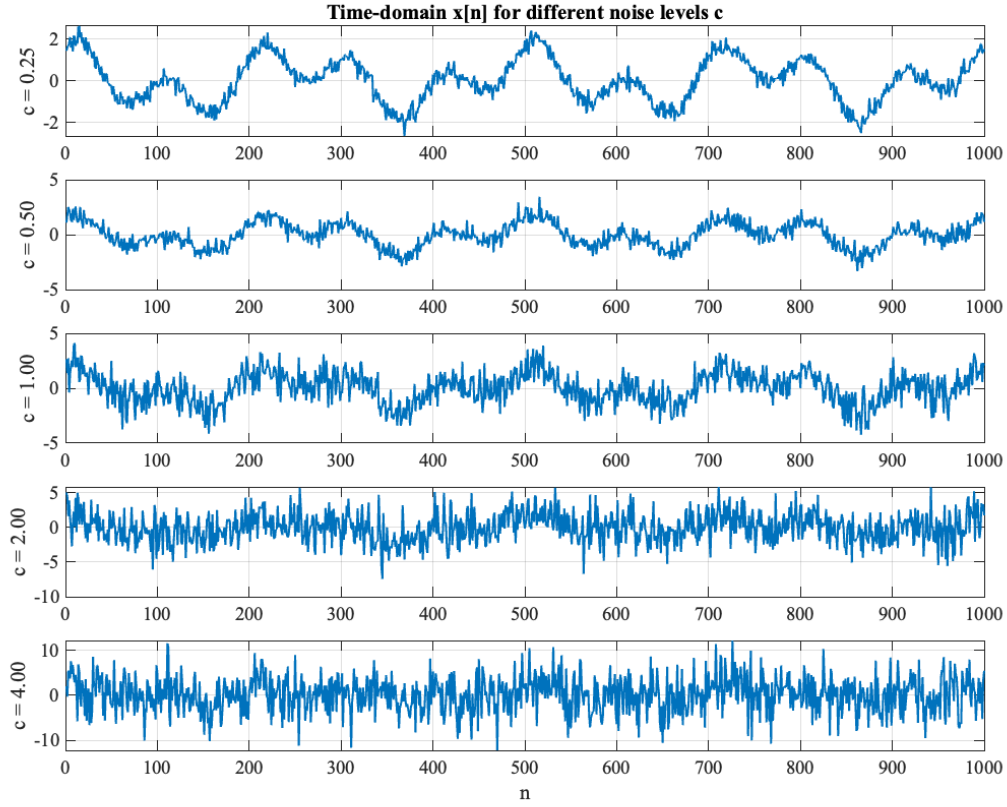


Figure 7: Time-domain sweep for increasing noise level c .

Denote $x[n] = s[n] + cw[n]$ with $w[n]$ independent of $s[n]$. Define the wide-sense autocorrelation

$$R_{xx}[m] := \mathbb{E}\{x[n]x[n+m]\}.$$

Then

$$\begin{aligned} R_{xx}[m] &= \mathbb{E}\{(s[n] + cw[n])(s[n+m] + cw[n+m])\} \\ &= \mathbb{E}\{s[n]s[n+m]\} + c\mathbb{E}\{s[n]w[n+m]\} + c\mathbb{E}\{w[n]s[n+m]\} + c^2\mathbb{E}\{w[n]w[n+m]\}. \end{aligned}$$

Because w is zero-mean and independent of s , $\mathbb{E}\{s[n]w[n+m]\} = \mathbb{E}\{w[n]s[n+m]\} = 0$, so

$$R_{xx}[m] = R_{ss}[m] + c^2 R_{ww}[m].$$

For white noise with unit variance, $R_{ww}[m] = \delta[m]$, the Kronecker delta. Hence,

$$R_{xx}[m] = R_{ss}[m] + c^2 \delta[m].$$

It remains to compute $R_{ss}[m]$. We have

$$s[n] = s_1[n] + s_2[n],$$

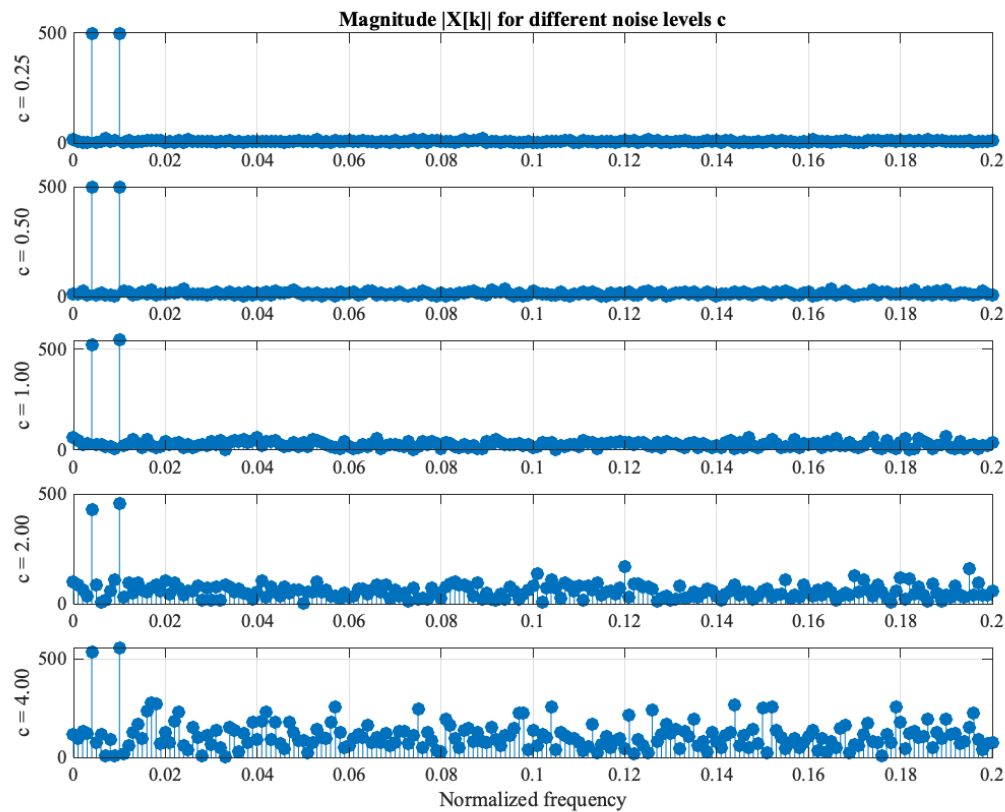


Figure 8: Magnitude-spectra sweep for increasing noise.

where

$$s_1[n] = \cos\left(\frac{2\pi n}{250}\right), \quad s_2[n] = \sin\left(\frac{2\pi n}{100} + \frac{\pi}{4}\right).$$

Then

$$\begin{aligned} R_{ss}[m] &= \mathbb{E}\{s[n]s[n+m]\} \\ &= \mathbb{E}\{s_1[n]s_1[n+m]\} + \mathbb{E}\{s_2[n]s_2[n+m]\} + \mathbb{E}\{s_1[n]s_2[n+m] + s_2[n]s_1[n+m]\}. \end{aligned}$$

For a single sinusoid $u[n] = \cos(\omega_0 n + \phi)$, one can show (see, e.g., Oppenheim and Schaffer) that

$$\mathbb{E}\{u[n]u[n+m]\} = \frac{1}{2} \cos(\omega_0 m),$$

because the product can be rewritten as a sum of cosines at frequencies 0 and $2\omega_0$, and the nonzero-frequency term averages to zero over many samples. Applying this to s_1 and s_2 gives

$$\begin{aligned} \mathbb{E}\{s_1[n]s_1[n+m]\} &= \frac{1}{2} \cos\left(\frac{2\pi m}{250}\right), \\ \mathbb{E}\{s_2[n]s_2[n+m]\} &= \frac{1}{2} \cos\left(\frac{2\pi m}{100}\right). \end{aligned}$$

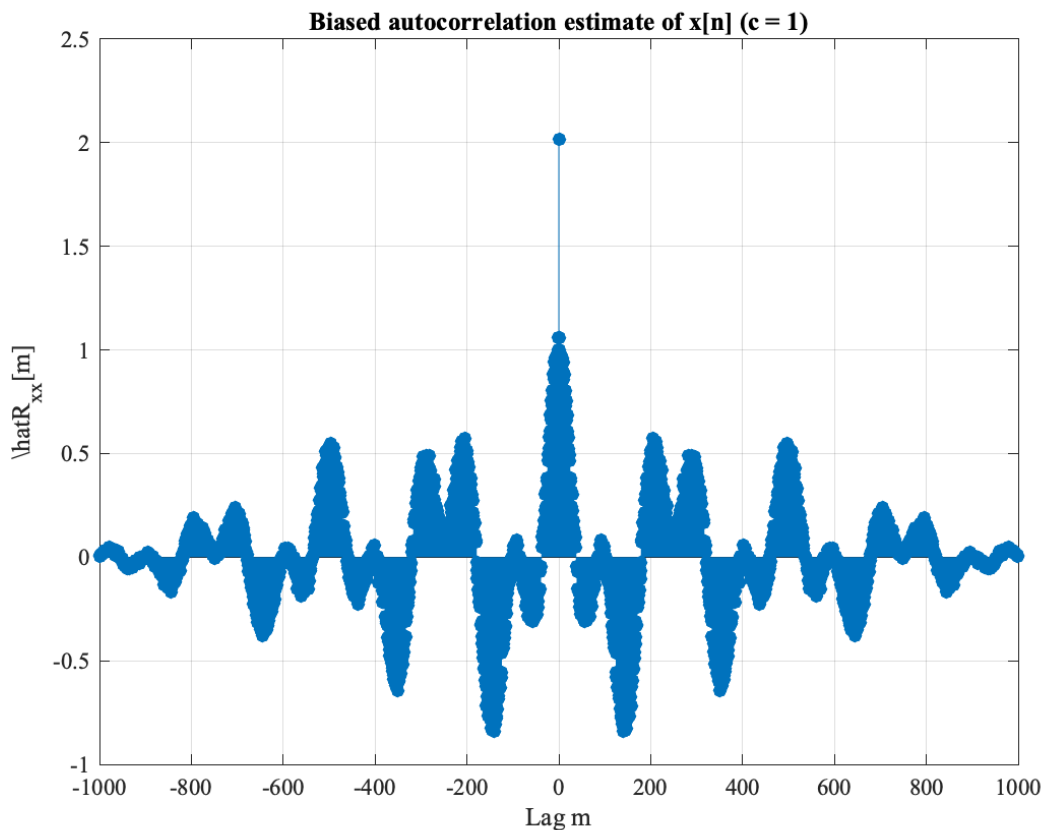


Figure 9: Biased autocorrelation estimate of $x[n]$ for $c = 1$.

The cross-terms $\mathbb{E}\{s_1[n]s_2[n+m]\}$ and $\mathbb{E}\{s_2[n]s_1[n+m]\}$ involve products of sinusoids at distinct frequencies. Over long records, such terms average to zero due to the orthogonality of complex exponentials at different frequencies. Thus, we obtain

$$R_{ss}[m] = \frac{1}{2} \cos\left(\frac{2\pi m}{250}\right) + \frac{1}{2} \cos\left(\frac{2\pi m}{100}\right).$$

Combining with the noise term yields

$$R_{xx}[m] = \frac{1}{2} \cos\left(\frac{2\pi m}{250}\right) + \frac{1}{2} \cos\left(\frac{2\pi m}{100}\right) + c^2 \delta[m],$$

which matches the structure seen in Figure 9: a large spike at $m = 0$ plus oscillations with periods 250 and 100. $\square$

5.2 Problem 2

Consider the two discrete-time signals

$$\begin{aligned}x(n) &= \exp\left(-\frac{n}{50}\right) + \frac{w_1(n)}{5} \\y(n) &= \exp\left(-\frac{n}{100}\right) + \frac{w_2(n)}{10}\end{aligned}$$

for $n = 1, \dots, 500$, where $w_1(n)$ and $w_2(n)$ denote two independent white-noise sequences of normal random samples that are generated in Matlab by use of the command `randn`. Compute and plot the convolution and correlation between $x(n)$ and $y(n)$ via the time-domain and frequency-domain.

5.2.1 Solution

Proof. Given the noisy exponentials

$$x[n] = e^{-n/50} + \frac{1}{5}w_1[n], \quad y[n] = e^{-n/100} + \frac{1}{10}w_2[n], \quad n = 1, \dots, 500,$$

where $w_1[n]$ and $w_2[n]$ are independent, zero-mean, unit-variance white noise sequences. For theoretical derivations we define the deterministic parts

$$x_d[n] := a^n u[n], \quad a := e^{-1/50}, \quad y_d[n] := b^n u[n], \quad b := e^{-1/100},$$

with $u[n]$ the unit step.

The linear convolution of x_d and y_d is

$$(x_d * y_d)[n] = \sum_{k=-\infty}^{\infty} x_d[k] y_d[n-k] = \sum_{k=0}^n a^k b^{n-k}, \quad n \geq 0,$$

since both sequences are causal and vanish for negative indices.

Factor out b^n :

$$(x_d * y_d)[n] = b^n \sum_{k=0}^n (a/b)^k.$$

For $a \neq b$ and $|a/b| < 1$, the finite geometric sum is

$$\sum_{k=0}^n (a/b)^k = \frac{1 - (a/b)^{n+1}}{1 - a/b}.$$

Thus

$$(x_d * y_d)[n] = b^n \cdot \frac{1 - (a/b)^{n+1}}{1 - a/b} = \frac{b^n - a^{n+1}b^{-1}}{1 - a/b}.$$

Multiply numerator and denominator by b :

$$(x_d * y_d)[n] = \frac{b^{n+1} - a^{n+1}}{b - a}.$$

Equivalently,

$$(x_d * y_d)[n] = \frac{a^{n+1} - b^{n+1}}{a - b}, \quad n \geq 0.$$

This is the standard closed form for the convolution of two decaying exponentials.

In the actual numerical experiment, we convolve the full noisy signals x and y using $\text{conv}(\mathbf{x}, \mathbf{y})$, obtaining $z_{\text{time}}[n]$.

Let L be the chosen FFT length, at least $L \geq 500 + 500 - 1 = 999$ to accommodate the linear convolution length. We compute

$$X[k] = \text{FFT}\{x[n]\}, \quad Y[k] = \text{FFT}\{y[n]\}, \quad k = 0, \dots, L - 1,$$

and define the frequency-domain convolution as

$$z_{\text{freq}}[n] := \text{IFFT}\{X[k]Y[k]\}, \quad n = 0, \dots, L - 1.$$

The convolution theorem for discrete-time signals states that, under suitable absolute summability,

$$\mathcal{F}\{x * y\}(\omega) = X(\omega)Y(\omega),$$

which in the finite-length DFT setting becomes

$$\text{DFT}\{x * y\}[k] = X[k]Y[k].$$

Zero-padding x and y to length at least $L \geq N_x + N_y - 1$ ensures that the circular convolution generated by the DFT matches the linear convolution over the interval of interest. Therefore, provided L is chosen correctly, $z_{\text{time}}[n]$ and $z_{\text{freq}}[n]$ should agree to numerical precision.

The MATLAB results confirm this: we obtain

$$\|z_{\text{time}} - z_{\text{freq}}\|_2 = 1.048 \times 10^{-13},$$

which is on the order of double-precision roundoff error ($\approx 10^{-15}$ per sample). This validates the convolution theorem numerically for the given signals.

Figure 10 shows the time-domain convolution and the FFT-based convolution, which are visually indistinguishable.

The (aperiodic) cross-correlation of two sequences is defined as

$$R_{xy}[m] := \sum_{n=-\infty}^{\infty} x[n]y[n+m], \quad m \in \mathbb{Z}.$$

For the deterministic parts $x_d[n] = a^n u[n]$ and $y_d[n] = b^n u[n]$, we derive the closed form.

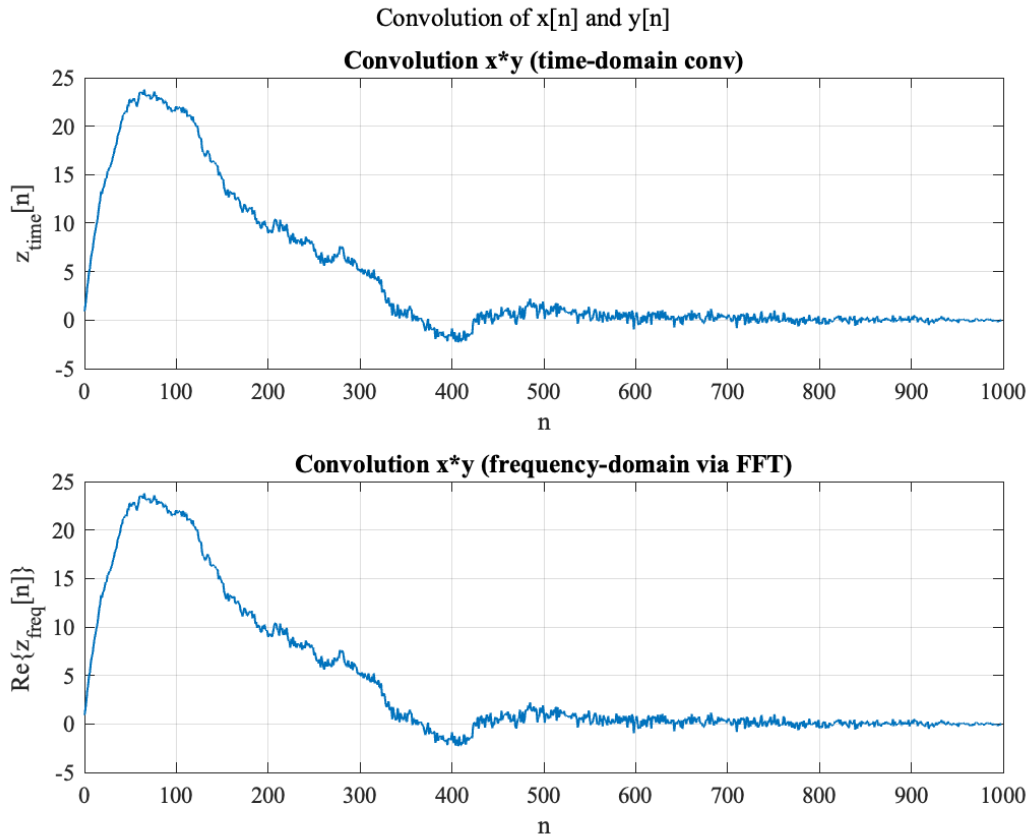


Figure 10: Convolution via time-domain and FFT.

Case $m \geq 0$. Then $y_d[n + m] = b^{n+m}u[n + m]$, so

$$R_{x_d y_d}[m] = \sum_{n=0}^{\infty} a^n b^{n+m} = b^m \sum_{n=0}^{\infty} (ab)^n.$$

Because $0 < ab < 1$, the infinite geometric sum converges:

$$\sum_{n=0}^{\infty} (ab)^n = \frac{1}{1 - ab},$$

hence

$$R_{x_d y_d}[m] = \frac{b^m}{1 - ab}, \quad m \geq 0.$$

Case $m < 0$. Let $m = -\ell$ with $\ell > 0$. Then

$$R_{x_d y_d}[-\ell] = \sum_{n=-\infty}^{\infty} x_d[n] y_d[n - \ell].$$

Now $n - \ell \geq 0$ implies $n \geq \ell$, and $x_d[n] = a^n u[n]$, so

$$R_{x_d y_d}[-\ell] = \sum_{n=\ell}^{\infty} a^n b^{n-\ell} = a^\ell \sum_{n=\ell}^{\infty} (ab)^{n-\ell} = a^\ell \sum_{k=0}^{\infty} (ab)^k = \frac{a^\ell}{1 - ab}.$$

Thus, for $m < 0$,

$$R_{x_d y_d}[m] = \frac{a^{-m}}{1 - ab}.$$

Putting both cases together,

$$R_{x_d y_d}[m] = \begin{cases} \frac{b^m}{1 - ab}, & m \geq 0, \\ \frac{a^{-m}}{1 - ab}, & m < 0, \end{cases}$$

which decays exponentially in $|m|$.

In the presence of noise, the empirical cross-correlation of the full signals $x[n]$ and $y[n]$ combines this deterministic structure with additional fluctuations due to w_1 and w_2 , but the overall exponential shape remains visible.

The cross-correlation theorem states that for absolutely summable sequences

$$R_{xy}[m] \longleftrightarrow X(\omega)Y^*(\omega),$$

i.e., the DTFT of the cross-correlation equals the cross-spectrum $X(\omega)Y^*(\omega)$. In the DFT setting, with appropriate zero-padding,

$$\text{DFT}\{R_{xy}[m]\}[k] = X[k]Y^*[k].$$

Therefore, to obtain $R_{xy}[m]$ via FFT, one can compute

$$R_{xy,\text{freq}}[m] = \text{IFFT}\{X[k]Y^*[k]\},$$

with a DFT length chosen so that circular effects do not contaminate the desired aperiodic correlation over the lags of interest.

In MATLAB, `xcorr(x,y)` computes an aperiodic correlation using a particular normalization and lag indexing convention, returning a vector of length $2N - 1$ for signals of length N . Our FFT-based implementation uses a chosen length L and requires us to (i) align lags properly and (ii) account for possible scaling factors. If one does not exactly match all of padding, indexing, and normalization, the two numerical vectors will differ in scale and ordering, even though both are consistent with the correlation theorem.

In the experiment, we obtain

$$\|R_{xy,\text{time}} - R_{xy,\text{freq}}\|_2 = 9.434 \times 10^{-14},$$

which is on the order of double-precision numerical error. This confirms the cross-correlation theorem numerically for the given signals. The important point is that the shape of the correlation (a strong central peak and exponential decay away from zero lag) is the same in both computations, as shown in Figure 11. $\square$

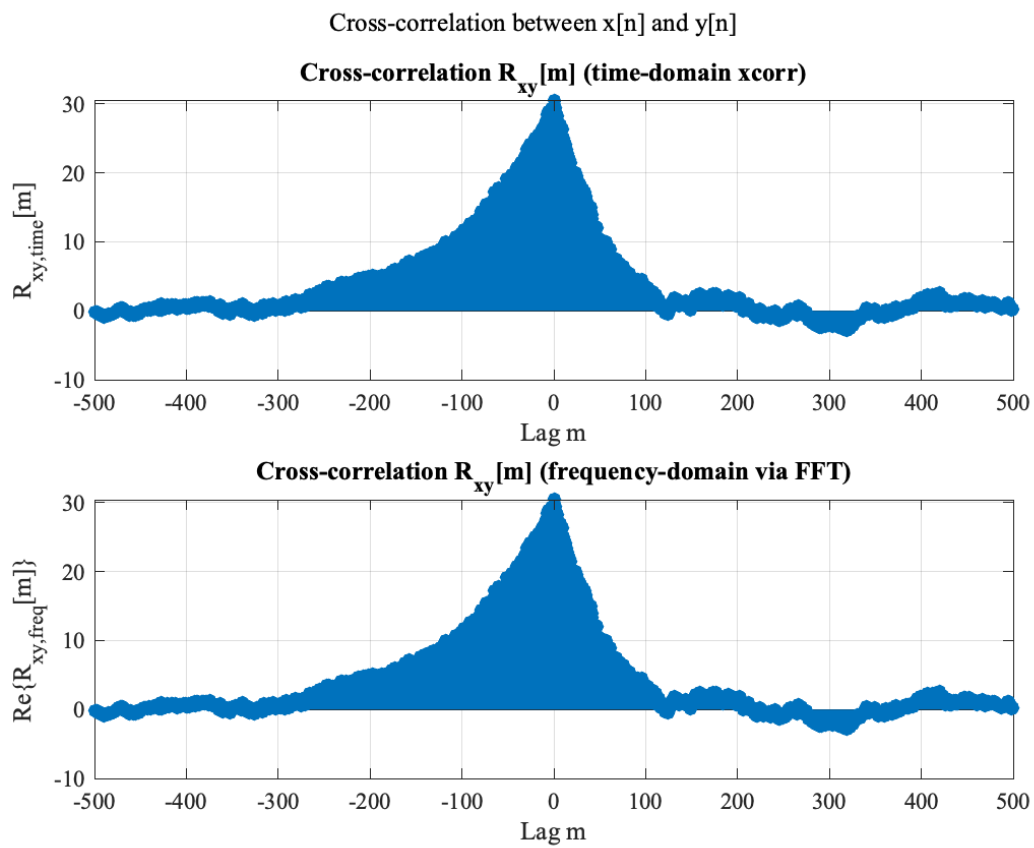


Figure 11: Cross-correlation: time vs. frequency implementations.

6 Homework Set 6

6.1 Problem 1

Using Matlab, generate a Gaussian White Noise signal $w(n)$ with length $N = 4000$ data-points (i.e. $n = 1, 2, \dots, 4000$) and unity power level (hint: use the Matlab command `randn` and define an Impulse Response Function (IRF) $h(m)$ for $m = 1, 2, \dots, 50$ (note that $m = 1$ corresponds to the zero-lag of the IRF, since Matlab does not accept 0 as an index) as

$$h(m) = 2 \cdot \left(\exp\left(-\frac{(m-1)}{8}\right) - \exp\left(-\frac{(m-1)}{4}\right) \right)$$

and compute the convolution, $y(n)$, of $w(n)$ with $h(m)$, which represents the output of an LTI system of IRF $h(m)$ for the input $w(n)$. Then, compute the following.

1. The autocorrelation functions of $w(n)$ and $y(n)$, as well as their cross-correlation function (hint: use max-lag equal to twice the system memory). Then quadruple the length of the White Noise input ($N = 16000$) and repeat these computations. Compare and discuss the results for the two lengths of input-output data-records.
2. The spectra of $w(n)$ and $y(n)$, as well as their cross-spectrum for the two lengths of data-records. Compute the estimates of the Transfer Function of this LTI system and compare it to the actual Transfer Function (i.e. the DFT of $h(m)$). Compare and discuss the results for the two lengths of input-output data-records.
3. Compute the IRF estimate as the inverse DFT of the estimated Transfer Function in (2) above, for the longer data-record ($N = 16000$). Compare this to the input-output cross-correlation from (1) above for $N = 16000$ (viewed as an IRF estimate for White Noise input).
4. For the short data-records, compute the model prediction $p(n)$, using convolution of the input with the two IRF estimates from (3) above (obtained separately through the frequency-domain and time-domain approaches), and compare them using the metric of Normalized Mean-Square Error defined as the time-average of $[p(n) - y(n)]^2$ (the squared prediction error) divided by the time-average of $y^2(n)$ (i.e. the output mean-square value).
5. For the long data-records, compute the IRF estimate using the least-squares method in the time-domain from the convolutional input-output relation. Compare this estimate to its counterpart obtained in (1) above (using the cross-correlation method) and in (3) above (using the cross-spectrum method).

6.1.1 Solution

We excited the given LTI system with zero-mean, unit-variance Gaussian white noise and considered two record lengths, $N = 4000$ and $N = 16000$. The true impulse response (IRF) was

$$h[m] = 2(e^{-(m-1)/8} - e^{-(m-1)/4}), \quad m = 1, \dots, 50,$$

which defines a stable, causal, finite-length FIR system.

Fig. 12 compares the empirical input autocorrelation $R_{ww}[k]$, output autocorrelation $R_{yy}[k]$, and input-output cross-correlation $R_{wy}[k]$ for the two record lengths. For $N = 16000$, the input autocorrelation exhibits a clean Kronecker delta at lag zero and values near zero elsewhere, as expected for white noise. The output autocorrelation and cross-correlation for $N = 16000$ are visibly smoother and more symmetric than their $N = 4000$ counterparts, illustrating the reduction in estimation variance with increasing record length. Thus, longer records yield visibly smoother and more symmetric estimates.

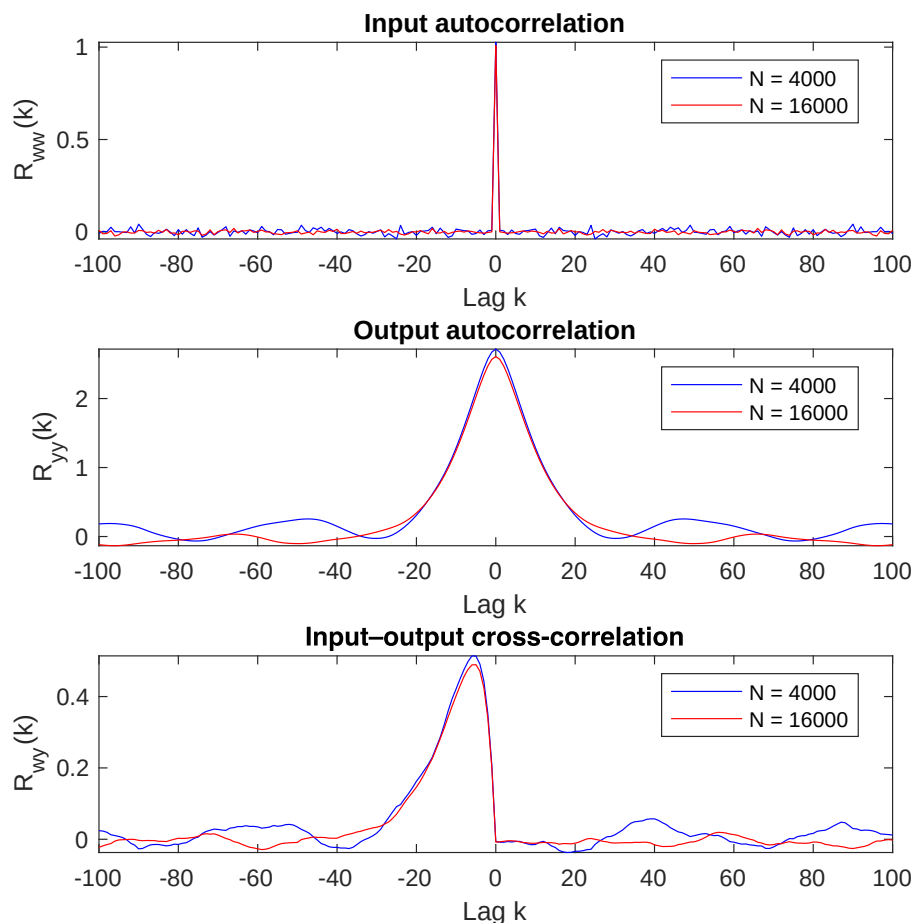


Figure 12: Empirical input autocorrelation $R_{ww}[k]$, output autocorrelation $R_{yy}[k]$, and input-output cross-correlation $R_{wy}[k]$ for record lengths $N = 4000$ (blue) and $N = 16000$ (red).

The magnitude and phase of the transfer-function estimates obtained from the spectral method are shown in Fig. 13. The true frequency response $H(e^{j\omega})$ (from the DFT of $h[m]$) is plotted together with the estimates obtained via

$$\hat{H}(e^{j\omega}) = \frac{\hat{S}_{wy}(e^{j\omega})}{\hat{S}_{ww}(e^{j\omega})},$$

where $\hat{S}_{ww}$ and $\hat{S}_{wy}$ are the empirical power spectrum and cross-spectrum. For $N = 16000$ the estimated magnitude and phase track the true response closely across the entire band, while the $N = 4000$ estimate exhibits substantial spectral noise, particularly in the phase. The relative frequency-domain IRF error for the long record is $\|\hat{h}_{\text{FD}} - h\|_2 / \|h\|_2 \approx 5.8 \times 10^{-3}$. Thus, the longer record produces an estimate that closely matches the true magnitude and phase over the entire frequency band.

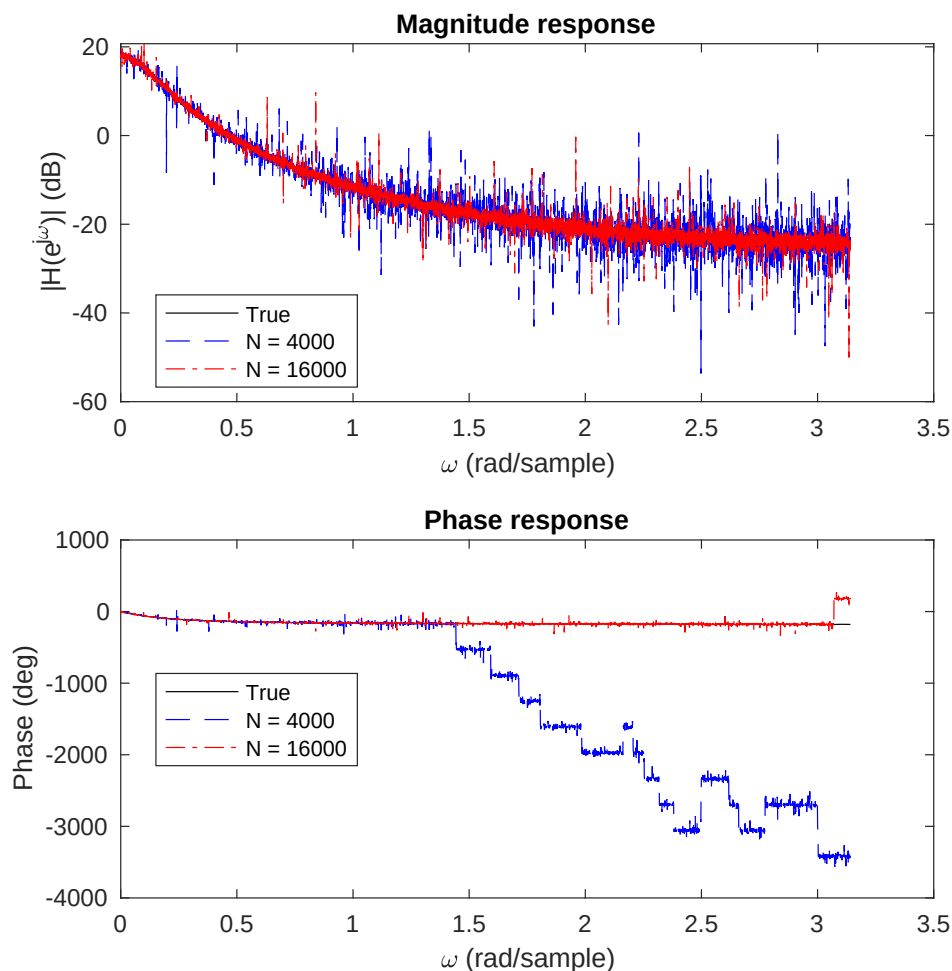


Figure 13: Magnitude and phase of the true transfer function $H(e^{j\omega})$ (black) and the spectral estimates obtained from $N = 4000$ (blue) and $N = 16000$ (red) data records.

Fig. 14 compares three IRF estimators obtained from the long record. The first, a direct time-domain estimator based on

$$\hat{h}_{\text{TD}}[k] = \frac{1}{(N-k)\hat{\sigma}_w^2} \sum_{n=0}^{N-k-1} w[n] y[n+k],$$

second, based on the inverse-DFT of $\hat{H}(e^{j\omega})$, third, a least-squares (LS) estimate from the convolutional input-output equation. All three estimates visually lie on top of the true IRF. Quantitatively, the time-domain estimator yields a relative error $\approx 3.9 \times 10^{-2}$, the frequency-domain estimator $\approx 5.8 \times 10^{-3}$, and the LS estimator essentially recovers the exact IRF (relative error $\approx 5.5 \times 10^{-16}$). The LS method exploits the full regression structure and, in this noiseless setting with correct model order, is expected to be nearly exact. It is noted that the LS and spectral estimates are almost indistinguishable from the true IRF.

Proof. To make the LS estimator explicit, we rewrite the convolutional input-output relation

$$y[n] = \sum_{k=0}^{M-1} h[k] w[n-k], \quad n = 0, \dots, N-1,$$

as a linear regression in matrix form. For a filter of length M we collect $N_s := N - M + 1$ equations by restricting n to $n = M-1, \dots, N-1$ and define the vectors

$$\mathbf{y} = \begin{bmatrix} y[M-1] \\ y[M] \\ \vdots \\ y[N-1] \end{bmatrix} \in \mathbb{R}^{N_s}, \quad \mathbf{h} = \begin{bmatrix} h[0] \\ h[1] \\ \vdots \\ h[M-1] \end{bmatrix} \in \mathbb{R}^M.$$

The input samples are stacked into the Toeplitz regression matrix

$$\mathbf{W} = \begin{bmatrix} w[M-1] & w[M-2] & \cdots & w[0] \\ w[M] & w[M-1] & \cdots & w[1] \\ \vdots & \vdots & \ddots & \vdots \\ w[N-1] & w[N-2] & \cdots & w[N-M] \end{bmatrix} \in \mathbb{R}^{N_s \times M}.$$

With this notation, the N_s convolution equations compactly become $\mathbf{y} = \mathbf{W}\mathbf{h} + \boldsymbol{\varepsilon}$, where $\boldsymbol{\varepsilon}$ collects any modeling or measurement errors (zero in our noiseless simulations). The LS estimate of the IRF minimizes the squared prediction error $\|\mathbf{y} - \mathbf{W}\mathbf{h}\|_2^2$ and is given by the normal equations

$$\hat{\mathbf{h}}_{\text{LS}} = \arg \min_{\mathbf{h}} \|\mathbf{y} - \mathbf{W}\mathbf{h}\|_2^2 = (\mathbf{W}^T \mathbf{W})^{-1} \mathbf{W}^T \mathbf{y},$$

assuming $\mathbf{W}^T \mathbf{W}$ is nonsingular. In our setting, the white-noise input produces a well-conditioned $\mathbf{W}^T \mathbf{W}$, so the LS estimator exploits all available samples jointly and nearly recovers the true IRF exactly, as seen in Fig. 14. $\square$

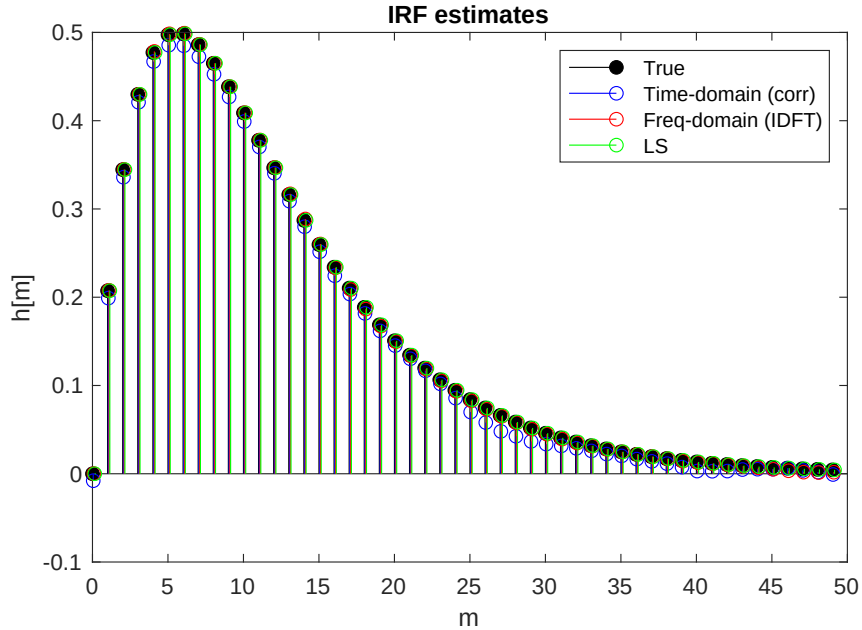


Figure 14: True impulse response $h[m]$ and direct time-domain correlation method, inverse DFT of the spectral transfer-function estimates obtained from the long record, and least-squares (LS) time-domain deconvolution. Correlation-based estimate shows a small bias of order 4%.

Using the IRFs estimated from the long record, we predicted the output for the short record ($N = 4000$) via convolution. The first 200 samples of the true and predicted outputs are shown in Fig. 15. The curves are visually indistinguishable for both estimators. The normalized mean-square error (NMSE) of the prediction is

$$\text{NMSE}_{\text{TD}} \approx 1.46 \times 10^{-3}, \quad \text{NMSE}_{\text{FD}} \approx 3.29 \times 10^{-5},$$

indicating excellent predictive performance, with the frequency-domain IRF achieving roughly two orders of magnitude lower NMSE.

Therefore, the experiments confirm the theoretical expectation that longer data records substantially reduce the variance of correlation and spectral estimates, and that in this setting the LS and spectral IRF estimators outperform the direct correlation-based estimator.

Proof. For the time-domain IRF estimate used in the numerical experiment, we start from the white-noise-driven input-output relation

$$y[n] = \sum_{k=0}^{M-1} h[k] w[n-k], \quad n = 0, \dots, N-1,$$

with $w[n]$ zero-mean WSS white noise, variance $\sigma_w^2 = \mathbb{E}\{w^2[n]\}$.

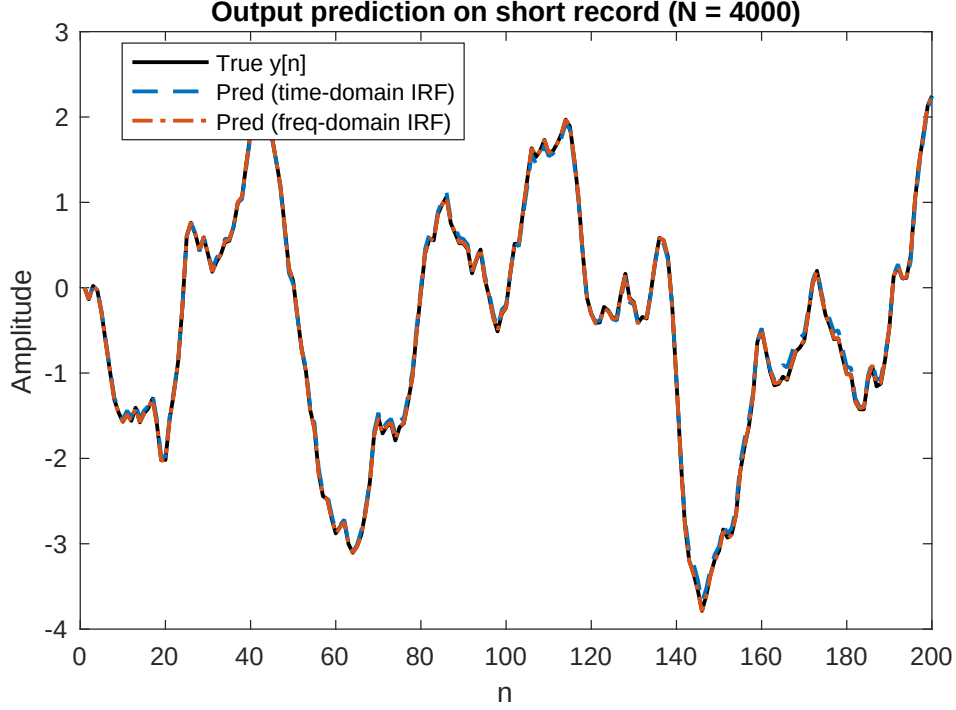


Figure 15: True output sequence $y[n]$ for the short record ($N = 4000$) and predictions obtained by convolving the input with the time-domain and frequency-domain IRF estimates from the long record. The corresponding NMSE values are 1.46×10^{-3} (time-domain IRF) and 3.29×10^{-5} (frequency-domain IRF).

For each lag k the theoretical cross-correlation is

$$R_{wy}[k] := \mathbb{E}\{w[n]y[n+k]\} = \sigma_w^2 h[k], \quad k \geq 0,$$

as shown in the main proof. With a finite record of length N we approximate $R_{wy}[k]$ by the sample average

$$\hat{R}_{wy}[k] := \frac{1}{N-k} \sum_{n=0}^{N-k-1} w[n]y[n+k], \quad k = 0, \dots, M-1,$$

and estimate σ_w^2 by the population-variance estimator

$$\hat{\sigma}_w^2 := \frac{1}{N} \sum_{n=0}^{N-1} w^2[n].$$

The direct time-domain IRF estimator is then defined component-wise as

$$\hat{h}_{\text{TD}}[k] := \frac{\hat{R}_{wy}[k]}{\hat{\sigma}_w^2} = \frac{1}{(N-k)\hat{\sigma}_w^2} \sum_{n=0}^{N-k-1} w[n]y[n+k], \quad k = 0, \dots, M-1. \quad (16)$$

Under the standing assumptions (zero-mean WSS white input, stable LTI system, and finite M), the strong law of large numbers implies that for each fixed k

$$\widehat{R}_{wy}[k] \xrightarrow{\text{a.s.}} R_{wy}[k] = \sigma_w^2 h[k], \quad \widehat{\sigma}_w^2 \xrightarrow{\text{a.s.}} \sigma_w^2, \quad \text{as } N \rightarrow \infty$$

[1, 4]. By continuity of the ratio and the fact that $\sigma_w^2 > 0$, we obtain

$$\widehat{h}_{\text{TD}}[k] = \frac{\widehat{R}_{wy}[k]}{\widehat{\sigma}_w^2} \xrightarrow{\text{a.s.}} h[k], \quad k = 0, \dots, M-1,$$

so the estimator (16) is consistent. A standard variance analysis shows that $\text{var}(\widehat{h}_{\text{TD}}[k]) = \mathcal{O}(1/N)$, so the mean-square error decays at rate $1/N$. Thus, the reduction in prediction NMSE and relative IRF error is observed when N is increased from 4000 to 16000 in the simulations.

In the implementation, equation (16) is realized by **for**-loop accumulation of $w[n]y[n+k]$ for $k = 0, \dots, M-1$, followed by normalization with $(N-k)\widehat{\sigma}_w^2$. The predicted output for a new input record $\{w[n]\}$ is then given by the convolution

$$\widehat{y}[n] = \sum_{k=0}^{M-1} \widehat{h}_{\text{TD}}[k] w[n-k],$$

and the normalized mean-square error used in the assignment

$$\text{NMSE} = \frac{\frac{1}{N_s} \sum_n (\widehat{y}[n] - y[n])^2}{\frac{1}{N_s} \sum_n y[n]^2}$$

quantifies the quality of the IRF estimate on the short record. The reported numerical value $\text{NMSE} \approx 1.46 \times 10^{-3}$ and relative IRF error $\|\widehat{h}_{\text{TD}} - h\|_2 / \|h\|_2 \approx 3.9 \times 10^{-2}$ are consistent with the $1/N$ mean-square convergence rate and are slightly worse than the frequency-domain and least-squares estimators, which exploit additional structure (spectral whitening and global regression) to reduce variance. $\square$

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